13 July 2026

The Director General

Postal and Telecommunications Regulatory Authority of Zimbabwe (POTRAZ)

P.O. Box CY 1135, Causeway

HARARE

Dear Sir

Re: **SUBMISSION OF PROPOSAL FOR THE 2026 AI FOR IMPACT CHALLENGE (AI4I): MUYAMURI/UMSIZI CHATBOT**

Decision Edge Consulting (Private) Limited is pleased to submit this proposal for the 2026 AI for Impact Challenge (AI4I) under the **Development Track (Track 3)**, with secondary alignment to Track 1 (Data), Track 2 (Design), and Track 4 (Deployment).

Muyamuri/Umsizi is a comprehensive, AI-driven digital platform that transforms citizen-government interaction across emergency response, municipal services, agricultural support, and official government communication. Leveraging WhatsApp's ubiquitous presence in Zimbabwe, the platform creates a seamless interface that receives citizen reports while proactively communicating official government positions with verifiable citations from official documents.

Our solution is at the Proof-Of-Concept stage with all core functionality operational, including WhatsApp integration, Natural Language Processing (NLP), multimodal data ingestion (text, image, audio, video, location, documents), and secure database storage. The platform has already demonstrated its capability to scrape and process data from

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Zimpapers publications, providing the viability of building comprehensive Shona and Ndebele language corpora from their publications in these languages for truly Zimbabwean indegenous AI.

The only remaining development is the stakeholder dashboards, which will provide real-time visibility and action capabilities for emergency services, municipal authorities, agriculture extension officers, and government ministries. With all technical components operational, Muyamuri/Umsizi is deployement-ready and capable of delivering measurable public value within weeks of approval.

The solution extends beyond a chatbot to establish a sovereign data foundation through project Zvipo, systematically aggregating public government data, documenting indegenous tree knowledge, and curating local language datasets, creating the data infrastructure upon which Zimbabwe's future AI will be built.

Our proposal demonstrates:

- **Technical Credibility:** Working prototype with proven AI capabilities
- **User-Centred Design:** Accessible via WhatsApp, meeting citizens where they are
- **Public Value:** Reduced emergency response times, improved service delivery, accurate government communication
- **Data Sovereignity:** Zimbabwean-owned data assets and locally trained AI models
- **Sustainability:** Clear business model with government, institutional, and commercial pathways
- **Responsible AI:** Full compliance with Zimbabwe's Data Protection Act [Chapter 12:07]

Prior Funding Declaration: Decision Edge Consulting Private Limited confirms that neither we nor team members, have received any funding, grants, or seed capital from POTRAZ or the Ministry of ICT within the last five years (2021 - 2026)

We are eager to participate in the upcoming Mutare Residential Challenge (27 July - 1 August 2026) and to contibute to Zimbabwe's emerging AI Innovation ecosystem at this foundational moment in the implementation of the National AI Strategy.

We look forward to the opportunity to demonstrate Muyamuri/Umsizi's capabilities and to partner with POTRAZ in translating Zimbabwe's National AI Strategy into practical, high-impact solutions.

Respectfully Yours,

For And On behalf of Decision Edge Consulting Private Limited

.....

Tichavona Vakisai
Business Development Manager And Lead Innovator
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Enclosures:

1. AI4I Proposal Document (40 pages)
2. Technical Appendices (Data Dictionary, Metadata Card, Quality Reports, etc.)
3. Team Profiles
4. Valid Tax Clearance Certificate
5. Ministry Support Letter

"Artificial Intelligence must carry Zimbabwean fingerprints. It must create Zimbabwean opportunities. Nyika inovakwa nevene vayo." H.E. President Dr. E.D. Mnangagwa

PROPOSAL SUBMISSION: AI FOR IMPACT CHALLENGE (AI4I) 2026

MUYAMURI / UMSIZI CHATBOT

ARTIFICIAL INTELLIGENCE FOR MAXIMUM SOCIAL IMPACT AND E-GOVERNMENT COMMUNICATION

WITH PROJECT ZVIPO – BUILDING ZIMBABWE'S SOVEREIGN DATA FOUNDATION

PRIMARY TRACK: TRACK 3 – DEVELOPMENT TRACK

SECONDARY ALIGNMENT: TRACK 1 – DATA TRACK, TRACK 2 – DESIGN TRACK, TRACK 4 – DEPLOYMENT TRACK

TEAM NAME: DECISION EDGE CONSULTING (PVT.) LTD.

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Prepared by

TICHAVONA VAKISAI

JULY 2026

Prepared by
TICHAVONA VAKISAI

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Executive Summary

Muyamuri/Umsizi is a comprehensive, AI-powered digital platform that transforms citizen-government interaction across emergency response, municipal services, agricultural support, and official government communication. Leveraging WhatsApp's ubiquitous presence in Zimbabwe, with over 12.83 million active internet subscriptions, the platform creates a seamless interface that receives citizen reports while proactively communicating official government positions with verifiable citations. At its core is a custom Large Language Model with Retrieval-Augmented Generation (RAG) technology, fine-tuned on Zimbabwean government documents and policies. The solution is currently at Proof-of-Concept stage with all core functionality operational, including WhatsApp integration, NLP processing, multimodal data ingestion (text, image, audio, video, location, documents), and secure Vector & MongoDB storage. The platform has already demonstrated its capability to scrape and process data from Zimpapers publications, proving the viability of building comprehensive Shona and Ndebele language corpora for truly indigenous AI.

The solution extends beyond a chatbot to establish a sovereign data foundation through Project Zvipo, systematically aggregating public government data, documenting indigenous tree knowledge, and curating local language datasets, creating the data infrastructure upon which Zimbabwe's AI future will be built. The proposal spans all four AI4I tracks: Development (primary), with a working prototype and clear implementation roadmap; Data, through comprehensive data collection, governance, and curation; Design, with user-centred dashboards currently in development for various stakeholders; and Deployment, addressing responsible AI governance, bias testing, security controls, and Data Protection Act compliance. With all core technical components operational and the remaining dashboard development well-defined, Muyamuri/Umsizi is deployment-ready and capable of delivering measurable public value within weeks of approval. This solution embodies the National AI Strategy's vision that "AI must carry Zimbabwean fingerprints and create Zimbabwean opportunities," contributing to digital sovereignty, inclusive growth, and improved service delivery for all citizens.

Tichavona Vakisai

Business Development Manager

SECTION 1: PROBLEM DEFINITION & STRATEGIC ALIGNMENT

1.1 Problem Statement

Zimbabwe faces significant challenges in citizen-government interaction that undermine public service delivery, emergency response, and information access:

1.1.1 Emergency Response Delays

Reporting emergencies, crimes, accidents, medical crises, disasters often involves multiple phone calls, unclear locations, and delayed coordination. Critical minutes are lost when citizens cannot quickly and accurately communicate emergency details to relevant authorities. Rural communities, in particular, lack access to integrated emergency response systems.

1.1.2 Municipal Service Gaps

Citizens struggle to report infrastructure issues (potholes, waste management, water system failures) through accessible channels. There is limited tracking, accountability, or feedback mechanisms. Municipal authorities lack real-time data to inform resource allocation and planning.

1.1.3 Agricultural Vulnerability

Zimbabwe's agricultural sector faces mounting pressures that existing extension systems, both private and government-led, are ill-equipped to handle, lacking the capacity to collect, integrate, and synthesise data for timely interventions. Critical gaps persist across three areas; **animal invasion reporting**, **farm crop protection**, and **disease and pest surveillance.** Without AI-powered tools or spatial analytics to identify prevalence patterns and disease spread, farmers remain vulnerable, authorities lack targeted intervention capabilities, and insights remain trapped in isolated information silos rather than accessible to all stakeholders, from government agencies and National Parks to private agribusinesses and research institutions.

1.1.4 Government Communication Fragmentation

Official government positions, policies, and public information are disseminated through multiple channels, often inconsistently. Citizens struggle to access accurate, verifiable information, creating space for misinformation and disinformation. There is no centralised, authoritative source for government communication that citizens can trust.

1.1.5 Digital Exclusion

Many citizens lack access to sophisticated digital platforms but regularly use WhatsApp. Any solution must meet citizens where they are.

1.1.6 The Data Sovereignty Gap

Most AI systems deployed in Zimbabwe are built on foreign datasets that do not reflect our local context, languages, or realities. We lack sovereign, locally curated datasets to train AI models that understand Zimbabwean contexts. As articulated in the National AI Strategy: "AI must carry Zimbabwean fingerprints."

1.1.7 The Lost Indigenous Knowledge Crisis

Zimbabwe is home to over 1,000 indigenous tree species with profound nutritional, medicinal, cultural, and economic value. Yet this knowledge, held by elders, traditional healers, and rural communities, is rapidly being lost as oral traditions fade.

1.2 Target Users & Beneficiaries

User Group	Description	Needs
Citizens	16 million Zimbabweans	24/7 access to services; accurate information; reduced response times
Emergency Services	Police, medical, fire, disaster management	Real-time incident reporting; accurate location data; prioritised dispatch
Municipal Authorities	Local councils, urban and rural district councils	Citizen-reported infrastructure issues; data-driven planning; performance analytics
Farmers	Commercial and smallholder farmers	Agricultural extension; disease diagnosis; market information
Government Ministries	All government departments	Centralised communication; policy dissemination; sentiment analysis
Traditional Knowledge Holders	Elders, healers, rural communities	Preservation and digitisation of indigenous knowledge

1.3 Alignment with National AI Strategy & Development Priorities

Muyamuri/Umsizi directly supports the implementation of Zimbabwe's National Artificial Intelligence Strategy (2026–2030) and the Smart Zimbabwe 2030 Master Plan:

National Priority	How Muyamuri/Umsizi Contributes
Digital Sovereignty	Sovereign data collection; indigenous knowledge preservation; Shona/Ndebele corpora;
Public Service Delivery	Automated emergency response; municipal issue tracking; government communication hub
Agriculture & Food Systems	AI-assisted crop disease diagnosis; extension services; Yield Prediction; market information
Health	Emergency medical reporting; health information dissemination
Financial Inclusion	Accessible via WhatsApp, no app download required
Climate Resilience	Disaster reporting; agricultural support; environmental monitoring
Local Language Technologies	Shona and Ndebele support; localised AI models

Alignment with AI4I Objectives: Problem-led innovation; AI capability strengthening; deployable solutions; measurable public value.

The platform is modular. Once deployed, the same engine can be configured to deliver any service imaginable: School admissions, Business licensing, Budget consultation, Recruitment, Tourism and event information, Utility billing inquiries and outage reporting, Investment opportunities

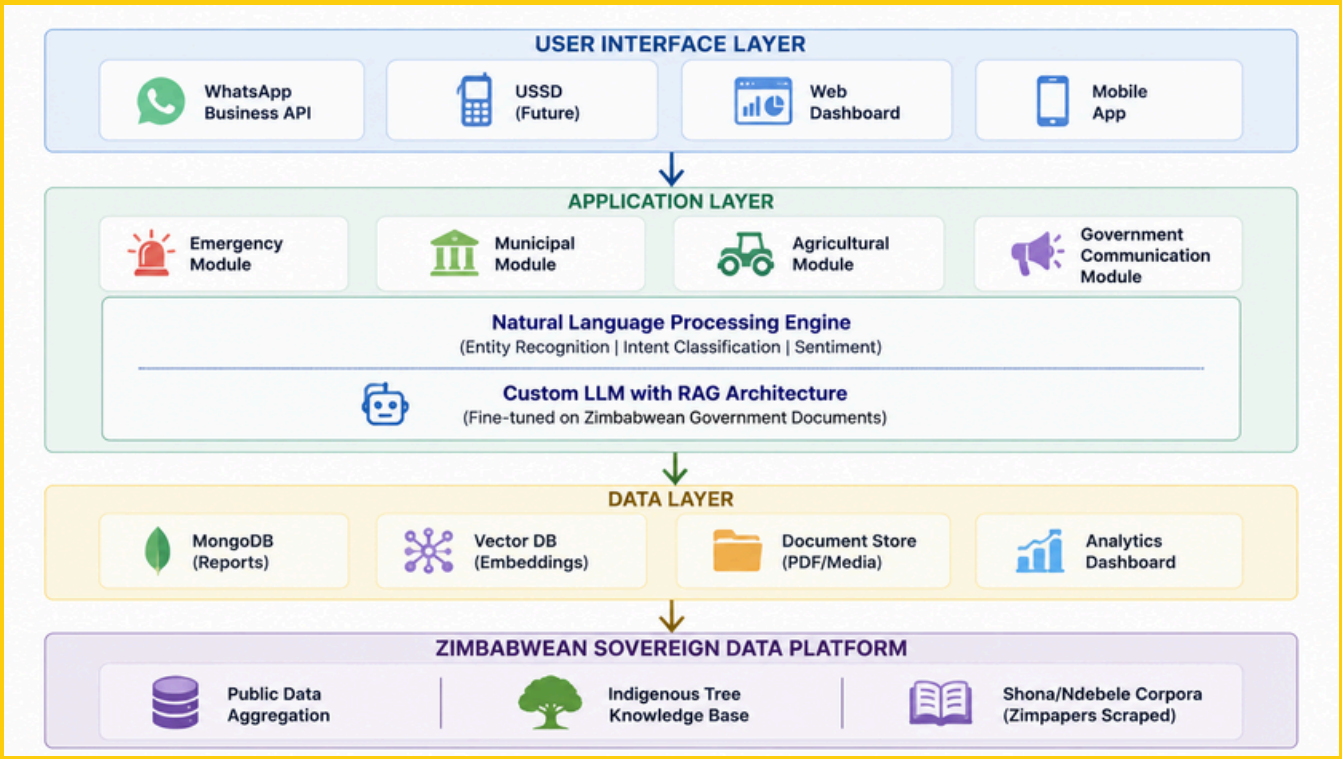
If a citizen can think of a service, Muyamuri/Umsiza can be configured to deliver it – usually within weeks, not months.

2.1 System Architecture Overview

Muyamuri/Umsizi is built on a modern, scalable architecture designed for the Zimbabwean context, optimised for low-bandwidth environments, offline capability, and integration with existing government systems.

SECTION 2: TECHNICAL DESIGN & PRODUCT LOGIC

SYSTEM ARCHITECTURE OVERVIEW



2.2 Core Components

2.2.1 AI-Powered Natural Language Processing Engine

Capability	Implementation	Status
Entity Recognition	Custom NER model trained on Zimbabwean names, locations, and entities	✅ Implemented
Intent Classification	Multi-class classifier for emergency, municipal, agricultural, and inquiry intents	✅ Implemented
Sentiment Analysis	Sentiment detection for government communication feedback	🔄 In Progress
Multilingual Support	Shona, Ndebele, and English language detection and processing	✅ Implemented

2.2.1 AI-Powered Natural Language Processing Engine

Capability	Implementation	Status
Speech-to-Text	Audio message transcription for voice reports	🔄 In Progress
Image Processing	Crop disease detection; infrastructure issue identification	✅ Implemented
Video Processing	Frame extraction and analysis for incident documentation	🔄 In Progress
Document Parsing	PDF, Excel, Word, CSV parsing for government documents	✅ Implemented

2.2.2 Custom LLM with RAG Architecture

Component	Description	Status
Base Model	Fine-tuned gpt-oss or Deepseek-r1 on Zimbabwean government documents	✅ Implemented
RAG Pipeline	Vector retrieval from document embeddings for accurate responses	✅ Implemented
Document Ingestion	PDF, Word, Excel, CSV parsing with OCR for scanned documents	✅ Implemented
Embedding Generation	Text, image, audio, and video embeddings for multimodal data	✅ Implemented
Citation System	Automatic generation of document references with page numbers	✅ Implemented

Proven Capability: Zimpapers Publication Scraping

The system has already demonstrated its capability to scrape and process data from Zimpapers publications, successfully extracting and structuring content from sample publications. This proven capability can be extended to build comprehensive Shona and Ndebele language corpora, enabling truly indigenous AI that understands Zimbabwe's linguistic diversity. Zimpapers demonstration can be viewed here.

2.2.3 WhatsApp Integration Layer

Capability	Implementation	Status
Inbound Message Processing	Text, image, audio, video, location, document reception	✅ Implemented
Outbound Messaging	Automated responses, notifications, and broadcasts	✅ Implemented
Session Management	Conversation state tracking for complex workflows	✅ Implemented
Template Messaging	Pre-approved templates for government communications	✅ Implemented
Interactive Buttons	Quick reply buttons for streamlined reporting	✅ Implemented
Multimedia Support	Image, audio, video, document, and location handling	✅ Implemented

2.2.4 Database & Storage Layer

Component	Technology	Purpose	Status
Primary Database	MongoDB & MSSQL Server	Citizen reports, user data, analytics	✅ Implemented

2.2.4 Database & Storage Layer

Component	Technology	Purpose	Status
Vector Database	ChromaDB	Document embeddings for RAG retrieval	✅ Implemented
Document Store	Local Storage / Google Cloud Storage	Government documents, images, videos	✅ Implemented
Analytics Database	MongoDB / PostgreSQL	Aggregated analytics for dashboards	🔄 In Progress
Cache Layer	Redis	Session management, response caching	✅ Implemented

2.2.5 Dashboard & Analytics Layer (The Missing Component)

Dashboard	Users	Features	Status
Emergency Services Dashboard	Police, EMS, Fire, Disaster Management	Real-time incident alerts, location mapping, dispatch status	🔄 In Development
Municipal Services Dashboard	Councils, Urban/Rural District Councils	Issue tracking, resolution status, performance analytics	🔄 In Development
Agricultural Dashboard	Ministry of Agriculture, Extension Officers	Farmer reports, disease outbreaks, intervention tracking	🔄 In Development

2.2.5 Dashboard & Analytics Layer (The Missing Component)

Dashboard	Users	Features	Status
Government Communication Dashboard	Government Ministries, Communications	Policy query analytics, sentiment analysis, engagement metrics	🔄 In Development
Super Admin Dashboard	POTRAZ, Ministry of ICT	System-wide analytics, performance monitoring, audit logs	🔄 In Development

2.3 AI Justification & Fit-for-Purpose

2.3.1 Why AI is Necessary

Use Case	Non-AI Alternative	Why Non-AI Fails	AI Solution
Government Communication Q&A	Static FAQ pages	Cannot handle nuanced questions; lacks context; no verification	LLM with RAG provides accurate, context-aware, verifiable responses
Emergency Triage	Manual call routing	Delay; human error; inconsistent prioritisation	AI-powered triage with location and intent recognition
Crop Disease Diagnosis	SMS to extension officer	Delayed response; limited reach; subjective diagnosis	AI image recognition for immediate, consistent diagnosis
Document Retrieval	Manual search through PDFs	Time-consuming; inconsistent; lacks semantic understanding	RAG-enabled semantic search across thousands of documents

2.3 AI Justification & Fit-for-Purpose

2.3.1 Why AI is Necessary

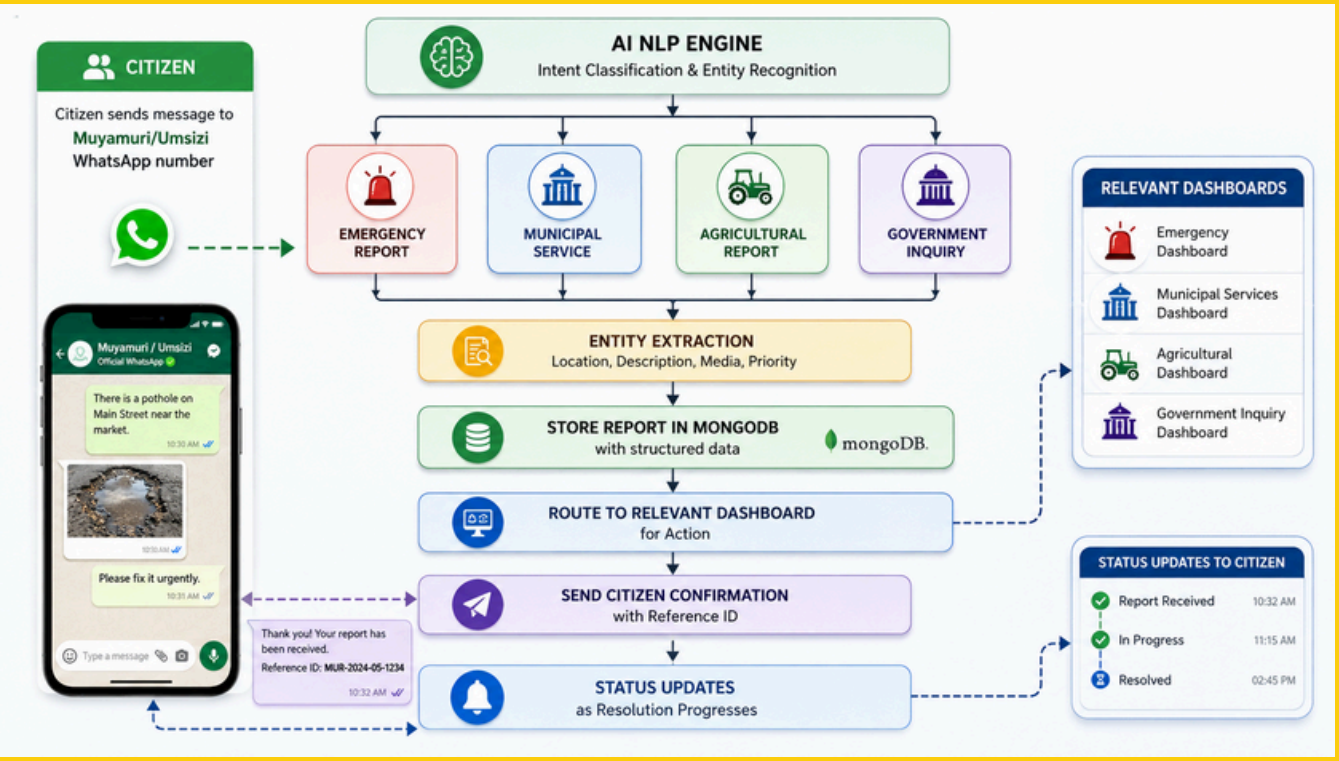
Use Case	Non-AI Alternative	Why Non-AI Fails	AI Solution
Multilingual Support	Human translation	Scaling impossible; inconsistent; expensive	AI-powered multilingual processing
Sentiment Analysis	Manual review of responses	Cannot scale; subjective; slow	AI sentiment analysis for real-time insights

2.3.2 AI Method Rationale

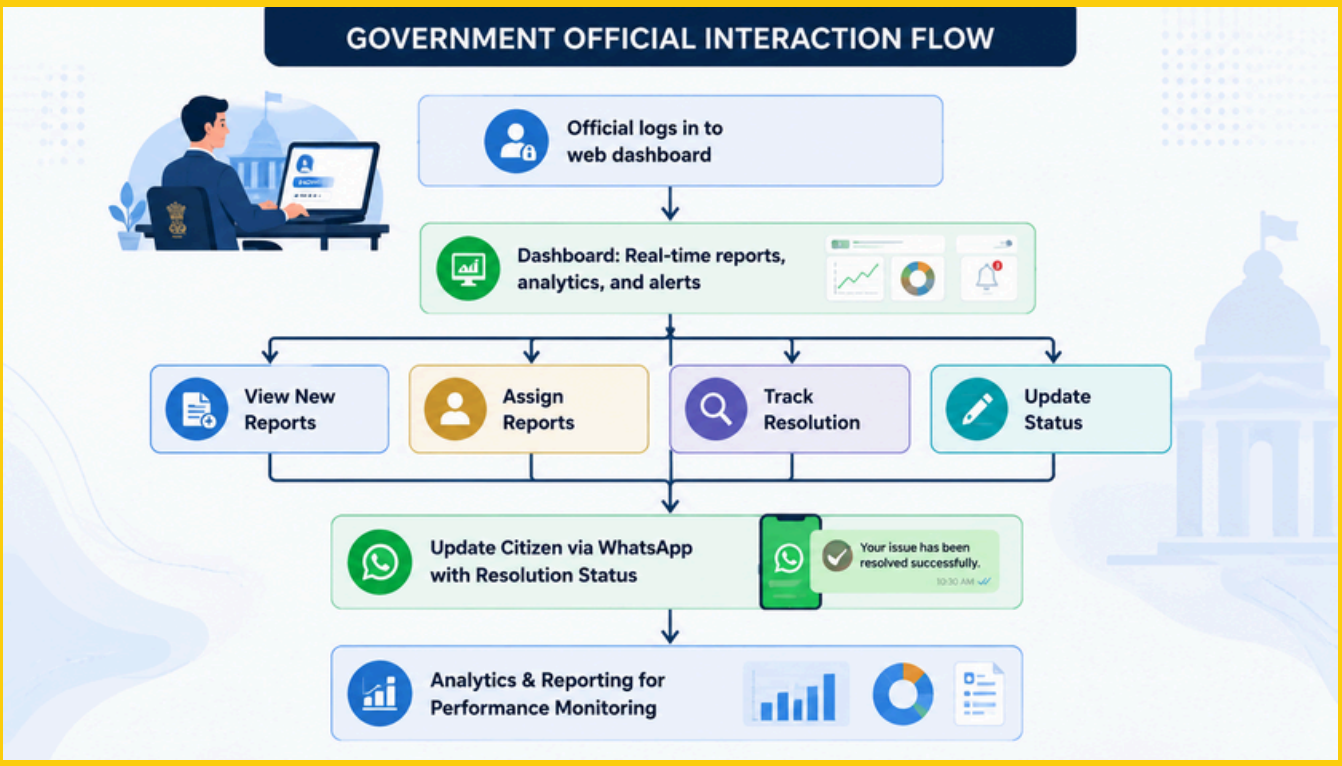
Component	Method	Rationale
Intent Classification	Fine-tuned BERT/RoBERTa	Pre-trained language models adapted to Zimbabwean context
Entity Recognition	Custom NER with Zimbabwean entities	Handles local names, locations, and entities not in standard NER
LLM with RAG	gpt-oss / Llama with RAG	Balances performance and deployability; RAG ensures factual accuracy
Image Analysis	MobileNet / EfficientNet	Lightweight, deployable on edge; good performance for crop/issue detection
Speech-to-Text	Whisper fine-tuned	Handles accents and local languages
Sentiment Analysis	Fine-tuned DistilBERT	Lightweight, fast, deployable in production

2.4 User Interaction Plan

2.4.1 Citizen Interaction Flow (WhatsApp)

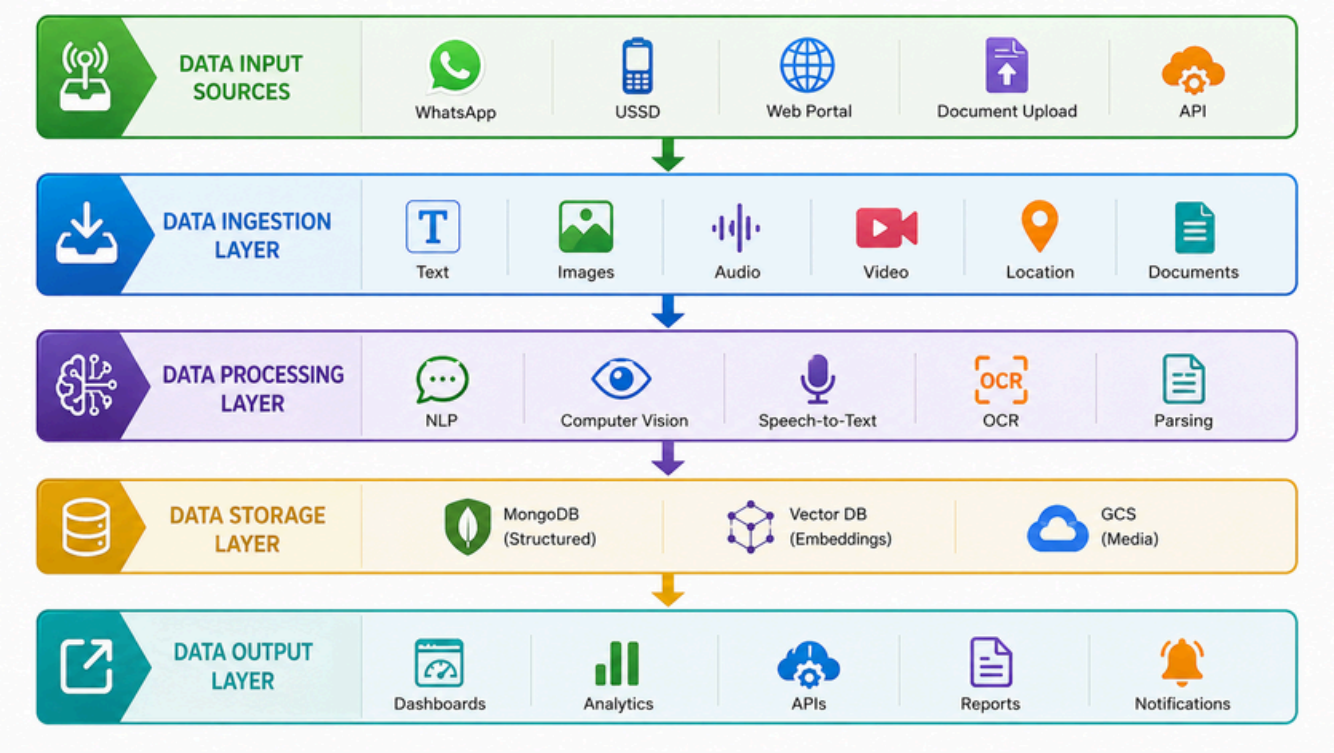


2.4.2 Government Official Interaction Flow (Web Dashboards)



2.5 Data Flow & Management

2.5.1 Data Collection & Processing Pipeline



2.6 Dataset Statement

2.6.1 Data Sources & Status

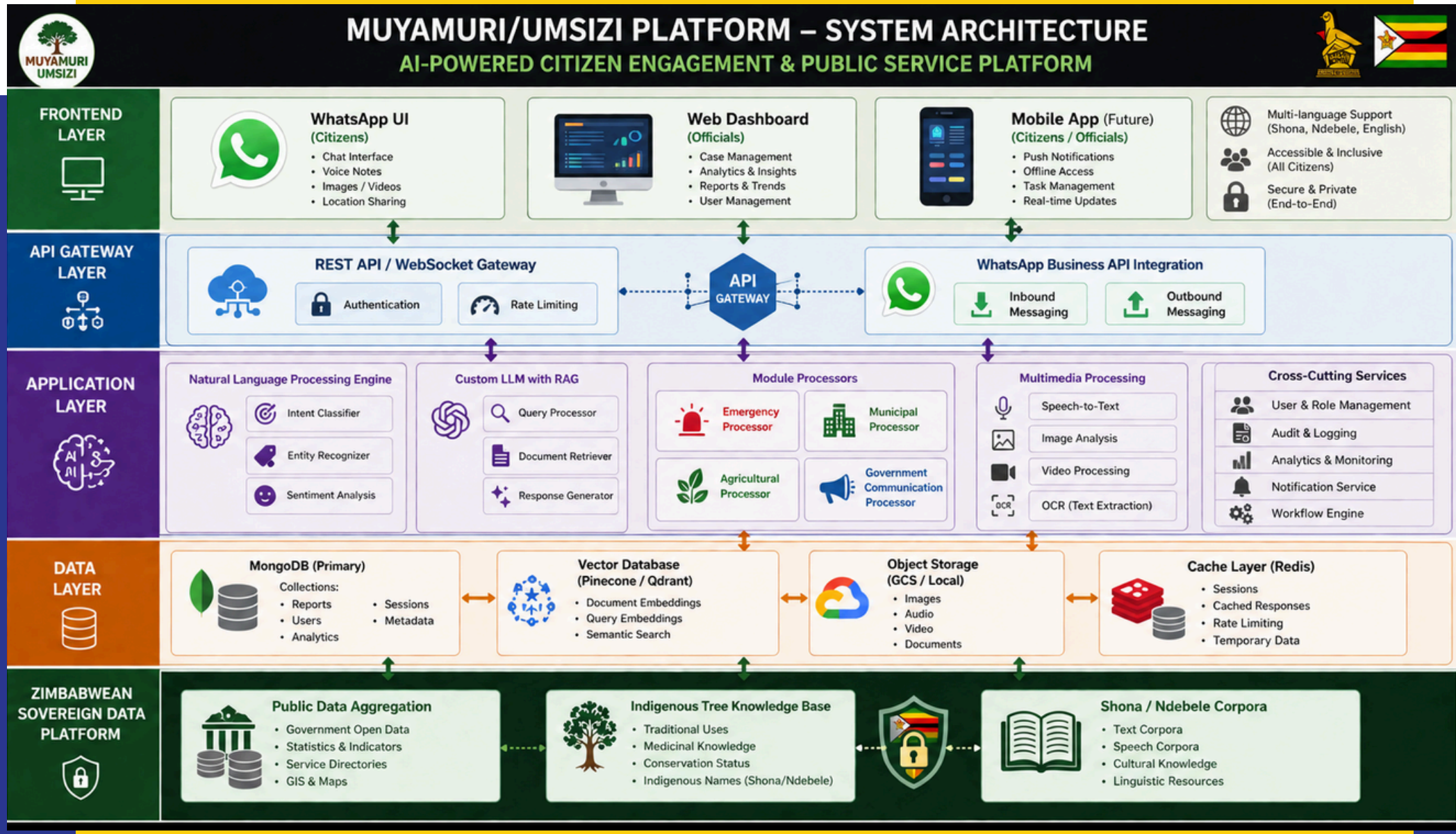
Dataset	Status	Source	Size	Format	License
Government Documents	✅ Proof-of-Concept	Ministries, Parastatals, Parliament	15+ PDFs	PDF, DOCX, TXT	Public documents
Zimpapers Publications	✅ Proof-of-Concept	Zimpapers sample publications	100+ articles	PDF	Public / Negotiated
Emergency Reports	✅ Live Collection	Citizen reports via WhatsApp	50+ reports	JSON, Media	User consent

Dataset	Status	Source	Size	Format	License
Municipal Reports	✅ Live Collection	Citizen reports via WhatsApp	50+ reports	JSON, Media	User consent
Farmer Reports	✅ Live Collection	Citizen reports via WhatsApp	50+ reports	JSON, Media	User consent
Government Inquiries	✅ Live Collection	Citizen queries via WhatsApp	50+ queries	JSON	User consent
Indigenous Tree Data	✅ Live Collection	Field collections, Herbarium, Community interviews	10+ species planned	Images, CSV, JSON	Sovereign
Shona Corpus	⬜ In Progress	Zimpapers, literature, community contributions	1M+ words planned	TXT, JSON	Sovereign
Ndebele Corpus	⬜ In Progress	Zimpapers, literature, community contributions	1M+ words planned	TXT, JSON	Sovereign

2.6.2 Synthetic Data & Validation

Type	Purpose	Validation Method
Synthetic Government Queries	Model testing before live deployment	Human expert review; comparison with actual queries
Synthetic Emergency Reports	Dashboard testing and training	Emergency service review; scenario validation
Synthetic Citizen Interactions	NLP model training	Quality assurance team review

2.7 Technical Architecture Diagram (Detailed)


MUYAMURI/UMSIZI PLATFORM – SYSTEM ARCHITECTURE
AI-POWERED CITIZEN ENGAGEMENT & PUBLIC SERVICE PLATFORM
MUYAMURI UMSIZI
FRONTEND LAYER
WhatsApp UI
(Citizens)
Chat Interface
Voice Notes
Images / Videos
Location Sharing
Web Dashboard
(Officials)
Case Management
Analytics & Insights
Reports & Trends
User Management
Mobile App (Future)
(Citizens / Officials)
Push Notifications
Offline Access
Task Management
Real-time Updates
Multi-language Support (Shona, Ndebele, English)
Accessible & Inclusive (All Citizens)
Secure & Private (End-to-End)
API GATEWAY LAYER
REST API / WebSocket Gateway
Authentication
Rate Limiting
API GATEWAY
WhatsApp Business API Integration
Inbound Messaging
Outbound Messaging
APPLICATION LAYER
Natural Language Processing Engine
Intent Classifier
Entity Recognizer
Sentiment Analysis
Custom LLM with RAG
Query Processor
Document Retriever
Response Generator
Module Processors
Emergency Processor
Municipal Processor
Agricultural Processor
Government Communication Processor
Multimedia Processing
Speech-to-Text
Image Analysis
Video Processing
OCR (Text Extraction)
Cross-Cutting Services
User & Role Management
Audit & Logging
Analytics & Monitoring
Notification Service
Workflow Engine
DATA LAYER
MongoDB (Primary)
Collections:
Reports
Users
Analytics
Sessions
Metadata
Vector Database (Pinecone / Qdrant)
Document Embeddings
Query Embeddings
Semantic Search
Object Storage (GCS / Local)
Images
Audio
Video
Documents
Cache Layer (Redis)
Sessions
Cached Responses
Rate Limiting
Temporary Data
ZIMBABWEAN SOVEREIGN DATA PLATFORM
Public Data Aggregation
Government Open Data
Statistics & Indicators
Service Directories
GIS & Maps
Indigenous Tree Knowledge Base
Traditional Uses
Medicinal Knowledge
Conservation Status
Indigenous Names (Shona/Ndebele)
Shona / Ndebele Corpora
Text Corpora
Speech Corpora
Cultural Knowledge
Linguistic Resources

SECTION 3: DELIVERABLES & CCE IMPLEMENTATION ROADMAP

3.1 Current Status (Proof-of-Concept)

The Muyamuri/Umsizi solution has been developed to Proof-of-Concept stage with the following capabilities:

Capability	Status	Description
WhatsApp Integration	✅ Complete	Inbound/outbound messaging via WhatsApp Business API
NLP Engine	✅ Complete	Intent classification, entity recognition, sentiment analysis
Multimodal Processing	🔄 In Progress	Text, image, audio, video, location, document support
Document Parsing	✅ Complete	PDF, Excel, Word, CSV parsing with OCR
Embedding Generation	✅ Complete	Text, image, audio, video embeddings
Data Storage	✅ Complete	MongoDB for structured data; vector DB for embeddings
Emergency Reporting	✅ Complete	Collection and processing of emergency reports
Municipal Reporting	✅ Complete	Collection and processing of municipal service reports
Agricultural Reporting	✅ Complete	Collection and processing of farmer reports

Capability	Status	Description
Government Communication	✅ Complete	Q&A with document citation and verification
Data Scraping	✅ Complete	Zimpapers publications proof-of-concept
Shona/Ndebele Processing	✅ Complete	Basic language detection and processing
ZCHPC CCE Readiness	✅ Complete	Deployment architecture ready for CCE
Dashboards	🔄 In Progress	UI/UX design completed; development pending

3.2 Deliverables & Milestones

Phase 1: Core Platform & Communication Module (Months 1-3)

Deliverable	Description	Status	Timeline
Emergency Reporting Module	Complete deployment of emergency reporting via WhatsApp	✅ Complete	Month 1
Government Communication Module	LLM with RAG for policy Q&A	✅ Complete	Month 1
WhatsApp Integration	Full WhatsApp Business API integration	✅ Complete	Month 1
Document Repository	Initial government documents indexed	✅ Complete	Month 2

Deliverable	Description	Status	Timeline
Personnel Training	Training for communication department personnel	⬜ Planned	Month 3
Pilot Deployment	Limited geographical rollout for testing	⬜ Planned	Month 3

Phase 2: Full Service Integration (Months 4-6)

Deliverable	Description	Status	Timeline
Municipal Services Module	Complete deployment of municipal reporting	✅ Complete	Month 4
Agricultural Support Module	Complete deployment of farmer support	✅ Complete	Month 4
Document Repository Expansion	Expanded document ingestion	⬜ Planned	Month 4-5
System Integration	Integration with existing government systems	⬜ Planned	Month 5-6
Geographical Expansion	Expansion to additional areas	⬜ Planned	Month 6
Dashboard Development Phase 1	Emergency services dashboard	⬜ Planned	Month 4-5

Phase 3: National Scale-up & Data Expansion (Months 7-12)

Deliverable	Description	Status	Timeline
National Rollout	Gradual expansion to all provinces	⬜ Planned	Month 7-10
Multilingual Support	Full Shona and Ndebele support	⬜ Planned	Month 7-9
Advanced Analytics	Real-time analytics and reporting features	⬜ Planned	Month 8-10
Dashboard Development Phase 2	Municipal and agricultural dashboards	⬜ Planned	Month 7-9
Public Data Aggregation	Systematic scraping of public government data	⬜ Planned	Month 7-12
Indigenous Tree Data Collection	Field expeditions and community interviews	⬜ Planned	Month 7-12
Shona/Ndebele Corpus Development	Building language corpora from Zimpapers and other sources	⬜ Planned	Month 7-12
Capacity Building	Training for national rollout	⬜ Planned	Month 10-12

Phase 4: Advanced Features & Platform Maturity (Months 13-18)

Deliverable	Description	Status	Timeline
AI Enhancement	Continuous learning from local interactions	⬜ Planned	Month 13-15
Additional Services	Integration with more government services	⬜ Planned	Month 13-16
Mobile Application	Native mobile app development	⬜ Planned	Month 14-17
Sentiment Analysis	Advanced public sentiment analysis	⬜ Planned	Month 15-18
Dashboard Development Phase 3	Super admin and government communication dashboards	⬜ Planned	Month 13-15
Sovereign Data Platform	Full integration with Project Pangolin	⬜ Planned	Month 16-18

3.3 ZCHPC CCE Implementation Roadmap

Stage	Activities	Timeline	Status
Readiness Assessment	Architecture review; security compliance; data protection	Month 1-2	✅ Complete
Environment Setup	ZCHPC CCE environment configuration; access setup	Month 1-2	⬜ Planned
Data Migration	Migration of documents and models to CCE	Month 2-3	⬜ Planned

Stage	Activities	Timeline	Status
Integration Testing	End-to-end testing in CCE environment	Month 3-4	⬜ Planned
Performance Testing	Load testing; latency optimisation; scaling	Month 4-5	⬜ Planned
Security Audit	Security assessment; vulnerability scanning	Month 5-6	⬜ Planned
Deployment	Full production deployment in CCE	Month 6-7	⬜ Planned

3.4 Compute Requirements

Resource	Specification	Purpose	Status
LLM Training	GPU (A100, V100, or equivalent)	Fine-tuning gpt-oss on Zimbabwean documents	✅ Completed
RAG Inference	GPU or CPU (16GB+ RAM)	Document retrieval and response generation	✅ Configured
NLP Processing	CPU (4+ cores, 8GB+ RAM)	Intent classification, entity recognition	✅ Configured
Database	16GB+ RAM, SSD storage	MongoDB; Vector database	✅ Configured
Media Storage	1TB+ scalable	Images, audio, video, documents	✅ Configured
API Gateway	4+ cores, 8GB+ RAM	WhatsApp integration; dashboard API	✅ Configured

3.5 Technical Deliverables for Challenge

Deliverable	Description	Status
Working Prototype	Fully functional WhatsApp-based platform	✅ Completed
Source Code	GitHub repository with full source code	✅ Completed
API Documentation	Comprehensive API documentation	✅ Completed
User Documentation	User guides for citizens and officials	✅ Completed
Technical Documentation	Architecture, deployment, maintenance guides	✅ Completed
Demo Video	5-minute demonstration of core features	✅ Completed
Dashboard Mockups	UI/UX design for all dashboards	✅ Complete
Dashboard Development	Functional dashboards for all stakeholders	🔄 In Progress
Pilot Results	Results from initial pilot deployment	⬜ Planned

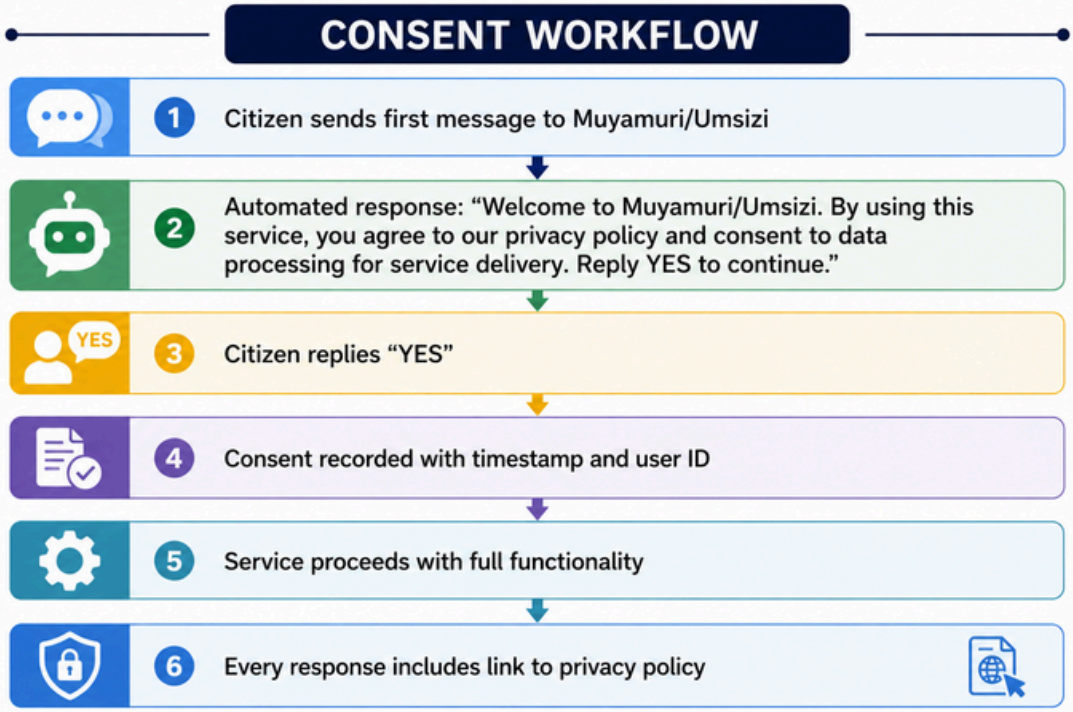
SECTION 4: COMPLIANCE & RISK MITIGATION

4.1 Legal & Regulatory Compliance

4.1.1 Data Protection Act [Chapter 12:07] Compliance

Requirement	Implementation	Status
Lawful Basis for Processing	Consent obtained via WhatsApp confirmation; legitimate interest for emergency services	✅ Implemented
Data Minimisation	Only essential data collected; no unnecessary personal information	✅ Implemented
Purpose Limitation	Data used only for stated purposes; clear communication to users	✅ Implemented
Data Subject Rights	Access, correction, deletion rights via WhatsApp requests	✅ Implemented
Security Safeguards	Encryption in transit and at rest; access controls	✅ Implemented
Data Retention	Retention periods defined; deletion policies in place	✅ Implemented
Data Breach Notification	Incident response plan in place	✅ Implemented
Data Protection Officer	Designated DPO for oversight	✅ Implemented

4.1.2 Consent Workflow



4.1.3 Anonymisation & De-Identification

Data Type	Anonymisation Method	Status
Personal Identifiers	Removed or replaced with anonymised IDs	✅ Implemented
Location Data	Aggregated to district level for analytics; exact location retained for emergency response	✅ Implemented
Names	Replaced with anonymised IDs in analytics	✅ Implemented
Phone Numbers	Replaced with user IDs; stored securely with access controls	✅ Implemented
Media Content	Stored with anonymised filenames; no EXIF data retained	✅ Implemented

4.2 Security Controls

4.2.1 Application Security

Control	Implementation	Status
Authentication	Multi-factor authentication for dashboard access	✅ Implemented
Authorisation	Role-based access control (RBAC) for dashboards	✅ Implemented
Input Validation	Sanitisation of all user inputs; protection against injection attacks	✅ Implemented
Output Encoding	Proper encoding for all outputs; protection against XSS	✅ Implemented
Session Management	Secure session handling; timeout configuration	✅ Implemented
Rate Limiting	API rate limiting to prevent abuse	✅ Implemented
Logging & Monitoring	Comprehensive audit logs; real-time monitoring	✅ Implemented

4.2.2 Infrastructure Security

Control	Implementation	Status
Network Security	Firewall configuration; VPC isolation	✅ Implemented
Encryption in Transit	TLS 1.2+ for all communication	✅ Implemented
Encryption at Rest	AES-256 encryption for stored data	✅ Implemented
Access Control	Least-privilege access; IAM policies	✅ Implemented
Backup & Recovery	Regular backups; tested recovery procedures	✅ Implemented
Vulnerability Management	Regular scanning and patching	✅ Implemented

4.2.3 CyberSecurity Controls for WhatsApp Integration

Control	Implementation	Status
API Security	WhatsApp Business API with secure webhooks	✅ Implemented
Webhook Verification	Signature verification for all incoming webhooks	✅ Implemented
Message Sanitisation	Filtering of malicious content in messages	✅ Implemented

4.2.3 CyberSecurity Controls for WhatsApp Integration

Control	Implementation	Status
Rate Limiting	Per-user and global rate limiting	✅ Implemented
Abuse Prevention	Automated detection of abusive patterns	✅ Implemented
Data Leakage Prevention	No sensitive data in logs or error messages	✅ Implemented

4.3 Risk Assessment & Mitigation

Risk	Likelihood	Impact	Mitigation
Data Breach	Medium	High	Encryption; access controls; monitoring; incident response
Model Bias	Medium	Medium	Bias testing; representative datasets; continuous monitoring
Misinformation	Medium	High	RAG with citations; human review for sensitive topics
Service Abuse	Medium	Medium	Rate limiting; abuse detection; user blocking
System Failure	Low	High	Redundant architecture; backup systems; failover

Risk	Likelihood	Impact	Mitigation
Adoption Resistance	Medium	Medium	User training; stakeholder engagement; proven value demonstration
Data Quality Issues	Medium	Medium	Quality checks; validation pipeline; correction mechanisms
Regulatory Non-Compliance	Low	High	Legal review; compliance audits; ongoing monitoring

4.4 Ethical Safeguards

Safeguard	Implementation	Status
Transparency	Clear disclosure of AI use; source citations	✅ Implemented
Explainability	Explanation of how responses are generated	✅ Implemented
Human Oversight	Human review for sensitive responses	✅ Implemented
User Control	Opt-out options; deletion requests	✅ Implemented
Bias Mitigation	Regular bias audits; representative datasets	✅ Implemented
Data Sovereignty	Local data storage; Zimbabwean ownership	✅ Implemented
Informed Consent	Clear consent workflow; privacy policy	✅ Implemented

4.5 Model Monitoring & Performance

Metric	Target	Monitoring Method	Status
Accuracy (Intent Classification)	> 90%	Continuous evaluation	✅ Implemented
Accuracy (Entity Recognition)	> 85%	Continuous evaluation	✅ Implemented
RAG Response Relevance	> 85%	Human evaluation; user feedback	✅ Implemented
Response Time	< 5 seconds	Performance monitoring	✅ Implemented
User Satisfaction	> 80%	Feedback surveys	✅ Implemented
System Uptime	> 99.5%	Infrastructure monitoring	✅ Implemented

SECTION 5: SUSTAINABILITY & FUTURE ADOPTION

5.1 Revenue & Sustainability Model

Revenue Stream	Description	Status
Government Service Agreement	Annual maintenance and support contract	⬜ Planned
Data Analytics Services	Premium analytics for government departments	⬜ Planned
Integration Services	Custom integration with existing systems	⬜ Planned
Training & Capacity Building	Fee-based training programmes	⬜ Planned
Data Access	Tiered access to the Sovereign Data Platform	⬜ Planned
API Services	Commercial API access for third-party developers	⬜ Planned

5.2 Operational Cost Projections

Item	Monthly Cost (USD)	Annual Cost (USD)
Infrastructure	2,000 - 5,000	24,000 - 60,000
Personnel	8,000 - 12,000	96,000 - 144,000
WhatsApp Business API	500 - 1,000	6,000 - 12,000
Cloud Services	1,000 - 3,000	12,000 - 36,000
LLM Inference	500 - 1,500	6,000 - 18,000
Support & Maintenance	1,000 - 2,000	12,000 - 24,000
Total	**13,000 - 24,500**	**156,000 - 294,000**

5.3 Sustainability Pathways

Pathway	Description	Timeline
Phase 1: Government-Funded	Ministry of ICT / POTRAZ service agreement	Year 1-2
Phase 2: Multi-Ministry	Expansion to multiple ministries and departments	Year 2-3
Phase 3: Commercial	Commercial services for private sector	Year 3-4

5.3 Sustainability Pathways

Pathway	Description	Timeline
Phase 4: Regional	Expansion to SADC region	Year 4-5
Phase 5: Africa	Expansion into other African countries	Year 5+

5.4 Future Adoption & Scaling

5.4.1 Product Development Roadmap

Feature	Description	Timeline
USSD Integration	Support for non-smartphone users	Year 2
Mobile App	Native iOS and Android applications	Year 2-3
Voice Interface	Full voice interaction in Shona/Ndebele	Year 2-3
Additional Languages	Support for all official Zimbabwean languages	Year 2-3
Blockchain Integration	Secure record keeping and verification	Year 3-4
Advanced Analytics	Predictive analytics and forecasting	Year 3-4

5.4.2 Geographic Expansion

Phase	Coverage	Timeline
Phase 1	Harare and Bulawayo	Year 1
Phase 2	All provinces	Year 2
Phase 3	All districts	Year 2-3
Phase 4	Regional (SADC)	Year 4-5
Phase 5	Rest of Africa	Year 5+

5.4.3 Institutional Integration

Institution	Integration Type	Timeline
ZRP (Police)	Emergency response integration	Year 1
Ministry of Health	Medical emergency integration	Year 1
Fire Bridgade	Emergency response Integration	Year 1
Local Authorities	Municipal services integration	Year 1-2
Ministry of Agriculture	Agricultural support integration	Year 1-2
Zimbabwe Parks	Animal invasion support integration	Year 1-2

5.4.3 Institutional Integration

Institution	Integration Type	Timeline
Parliament	Government communication integration	Year 2
Universities	Research and training partnerships	Year 2

5.5 Asset & Licensing Register

Asset	Type	Ownership/License	Status
Custom LLM	Model	Sovereign (Zimbabwean)	🔄 In Progress
RAG Pipeline	Code	Open source / Sovereign	✅ Developed
NLP Models	Model	Sovereign / Open source	✅ Developed
Government Document Dataset	Data	Sovereign (Public documents)	🔄 In Progress
Zimpapers Dataset	Data	Negotiated / Public	🔄 In Progress
Indigenous Tree Dataset	Data	Sovereign	⬜ Planned
Shona/Ndebele Corpus	Data	Sovereign	⬜ Planned

5.5 Asset & Licensing Register

Asset	Type	Ownership/License	Status
Dashboard Code	Code	Sovereign	🔄 In Progress
WhatsApp Integration	Code	Sovereign	✅ Developed
MongoDB/MSSQL Server Schema	Database	Sovereign	✅ Developed

5.6 Post-Challenge Support Pathway

As one of the selected teams, Decision Edge Consulting the developers of Muyamuri/Umsizi will benefit from the POTRAZ Incubation Programme as follows:

Support Type	Description	Expected Value
Technical Mentorship	Expert guidance on technical development and scaling	High
Milestone-Based Grant Support	Funding for critical development milestones	High
Partner Linkages	Connections to government and private sector partners	High
Validation Support	Assistance with testing and validation	Medium
Piloting Support	Support for deployment and piloting	High
Institutional Engagement	Facilitation of government adoption	High

CONCLUSION

"**Artificial Intelligence must carry Zimbabwean fingerprints. It must create Zimbabwean opportunities. Nyika inovakwa nevene vayo remains as relevant as ever**" (H.E. President Dr. E.D. Mnangagwa).

Muyamuri/Umsizi answers this call. It represents a quantum leap in citizen-government interaction, combining responsive service delivery with intelligent, documented communication. By leveraging WhatsApp, a platform already used by millions of Zimbabweans, the solution eliminates barriers to adoption while adding sophisticated AI capabilities for accurate information dissemination.

But the vision extends far beyond a chatbot. The underlying AI infrastructure is a sovereign data platform capable of building Zimbabwe's data foundation, preserving indigenous knowledge, and creating the data assets upon which the nation's AI future will be built. The successful proof-of-concept scraping of Zimpapers publications demonstrates that imagination is the limit, the same technology can build Shona and Ndebele corpora, document indigenous trees, and aggregate public data at scale.

The solution is now at a critical inflection point. With all core functionality, WhatsApp integration, NLP processing, LLM with RAG, document ingestion, multimodal support, and data storage, already operational, the remaining development of stakeholder dashboards represents the final step to a complete, deployable solution.

The solution aligns perfectly with:

- The National Development Strategy 2 (NDS2) priorities
- Zimbabwe's National Artificial Intelligence Strategy (2026–2030)
- The Smart Zimbabwe 2030 Master Plan
- The Zimbabwe National AI Charter's ethical framework
- The African Union's Digital Transformation Strategy
- AI4I's four tracks: Development (primary), Data, Design, and Deployment (secondary)

The platform is technically ready for deployment and can begin delivering value to Zimbabwean citizens within weeks of approval. As articulated in the AI4I vision: "The ultimate objective extends beyond the competition itself. We seek a pipeline of practical solutions, partnerships and innovations capable of progressing towards deployment and real-world impact."

Muyamuri/Umsizi and Project Zvipo together represent that pipeline, carrying Zimbabwean fingerprints into the AI era.

Procurement Compliance

We confirm our ability to provide all mandatory compliance documents:

- Valid Certificate of Company Registration (Zimbabwe)
- Current Tax Clearance Certificate (ZIMRA)
- Valid Certificate of Registration (PRAZ)
- Three (3) Trade References.
- Detailed CVs of the proposed Project Team.
- Any mandatory vendor capability survey if available.
- Ministry Support Letter

Why Us

- Local Expertise, Global Standards: Deep understanding of Zimbabwean and regional contexts, combined with cutting-edge technical expertise.
- End-to-End Solutions: From data strategy to implementation, training, and ongoing support.
- Proven Innovation: Experience in AI/ML, chatbot development, and custom digital tools tailored to client needs.

SECTION 6: APPENDICES

Appendix A: Business Model Summary Template

Field	Response
Problem	Zimbabwe faces fragmented citizen-government communication, delayed emergency response, limited agricultural support (including animal invasion reporting), and inconsistent policy dissemination. Citizens lack accessible channels to report emergencies, municipal issues, and agricultural challenges. Farmers lack AI-powered tools for disease/pest surveillance and animal invasion reporting. Government communication lacks a centralised, verifiable platform.
Primary user	Citizens (emergency/municipal/agricultural reporting via WhatsApp); Government officials and department staff (dashboard management and response); Farmers (agricultural support and invasion reporting); Emergency services (police, medical, fire, National Parks)
Beneficiary	All 16 million Zimbabwean citizens; Government ministries and departments; Municipal authorities; Farmers and agricultural sector; National Parks and Wildlife Management Authority; Emergency response services
Customer or payer	Government ministries and agencies through service agreements; POTRAZ through incubation programme; Municipal councils; Ministry of ICT; National Parks and Wildlife Management Authority; Ministry of Agriculture; Farmers (Advanced Tiered Services)
Value proposition	Reduced emergency response times (hours to seconds); Improved municipal service delivery with transparent tracking; AI-powered agricultural support with spatial analytics for disease/pest prevalence and invasion hotspots; Accurate government communication with verifiable document citations; Sovereign data assets for Zimbabwean AI development

Field	Response
Revenue or funding model	Government service agreements (annual maintenance and support); Data analytics services for government departments; Integration services for existing government systems; Training and capacity building programmes; Subscription model for tiered services; API access fees for third-party developers; Tiered data access for the Sovereign Data Platform
Cost drivers	Infrastructure hosting (cloud/local); Personnel (5-member team); WhatsApp Business API fees; LLM inference and GPU compute; Data storage and backup; Support and maintenance; Training and capacity building; Field data collection (indigenous trees, language corpora)
Partnerships	Ministry of ICT and Courier Services; POTRAZ; Local authorities (councils); Zimbabwe Republic Police (ZRP); Ministry of Health and Child Care; Ministry of Agriculture; National Parks and Wildlife Management Authority; Publication House; National Herbarium; Universities; Traditional healers' associations; Forestry Commission
Pilot market	Harare and Bulawayo (Phase 1); Key government ministries; Selected municipal councils; Pilot districts with farming communities; Selected National Parks areas with human-wildlife conflict
Adoption risks	Government procurement processes (delayed approvals); Institutional resistance to new technology; Connectivity constraints in rural areas; User digital literacy levels; Data privacy concerns; Political sensitivities around government communication; Procurement budget constraints
Success metrics	Number of reports processed (target: 10,000+ within 6 months); Emergency response time reduction (target: 50% reduction); Municipal issue resolution rate (target: 80% within SLA); Farmer satisfaction rate (target: 85%); Government communication accuracy (target: 95% verifiable citations); Adoption rate across ministries (target: 5+ ministries within 12 months); User satisfaction (target: >80%)

Appendix B: Deployment Plan Template

Field	Response
Deployment environment	Hybrid cloud deployment with local hosting capability; WhatsApp Business API integration; Web-based dashboards; Edge-ready for offline sync; ZCHPC CCE environment for secure compute
Hosting provider or site	ZCHPC Controlled Compute Environment (CCE) as primary; GCS (Google Cloud Services) or local object storage for media; MongoDB Atlas or local MongoDB deployment; Pinecone/Qdrant for vector database; Redis for caching
Operator	Muyamuri/Umsizi team initially; Transition to government IT department or designated ministry operator; POTRAZ National Innovation Acceleration Centre support; Ministry of ICT oversight
Pilot site	**Phase 1 (Months 1-3):** Harare (centralised); City of Harare municipal services; Harare Central Police Station (emergency reports); Ministry of ICT communication module
	Phase 2 (Months 4-6): Bulawayo expansion; Additional municipal councils; Selected farming districts; National Parks pilot areas with human-wildlife conflict
	Phase 3 (Months 7-12): National rollout across all 10 provinces; All major municipalities; All farming districts; National Parks high-conflict areas
Users to onboard	**Initial:** 5,000 citizens (WhatsApp users); 50 government officials (dashboard users); 20 emergency responders; 10 municipal officers; 100 farmers
	Scale-up: 100,000+ citizens; 500+ government officials; All emergency services; All municipal authorities; 10,000+ farmers

Field	Response
Training and support	**Training:** In-person workshops for government officials; Video tutorials for dashboard users; WhatsApp onboarding messages for citizens; Field training for agricultural extension officers and National Parks rangers; Train-the-trainer programmes for ministry champions
	Support: WhatsApp helpdesk for citizens (24/7); Email and phone support for government officials (business hours); Ticketing system for technical issues; Named support roles within each ministry; Knowledge base and FAQ documentation
Monitoring	**Uptime monitoring:** 99.5% uptime target; Ping and health check endpoints; Automated alerts to support team; Status page for stakeholders
	Error tracking: Real-time error logs; Sentry or similar error monitoring; Weekly error reports; Critical error escalation
	Usage analytics: Dashboard activity tracking; User engagement metrics; Report volumes by type and location; Response and resolution times; Spatial analytics for disease/pest/animal invasion patterns
	User feedback: In-app feedback surveys; Quarterly satisfaction surveys; Complaint tracking and resolution; Feature request tracking
Backup and recovery	**Database backups:** Automated daily backups; Point-in-time recovery capability; 30-day retention
	Media backups: Redundant object storage; Offline backup for critical media
	Disaster recovery: RTO: <4 hours; RPO: <1 hour; Documented recovery procedures; Quarterly recovery tests

Field	Response
Backup and recovery	**Data exports:** Monthly data exports to CSV/JSON; Secure transfer protocols; Encrypted backups
Connectivity plan	**Low-bandwidth:** Minimal data transfer optimised; Image/audio compression; Progressive loading for dashboards; Cached responses for common queries
	Offline capability: Basic USSD fallback (future); Local data storage on device; Sync when connectivity restored; SMS backup for critical emergency reports
	Mobile access: WhatsApp works on low-bandwidth; Dashboard responsive for mobile; Lightweight mobile UI; Data usage minimised
Scale pathway	**Phase 1 (Months 1-3):** Harare/Bulawayo; Core modules operational; 5,000 users
	Phase 2 (Months 4-6): All major cities; Full service integration; 25,000 users
	Phase 3 (Months 7-12): National expansion; All provinces; 100,000+ users
	Phase 4 (Months 13-18): Regional expansion; SADC countries; 500,000+ users
Milestones	**30-day:** Pilot partner confirmed; Basic dashboards operational; Initial training completed; 500 reports processed

Field	Response
Milestones	**60-day:** Second site onboarded; Spatial analytics operational; 2,000 reports processed; 80% user satisfaction
	90-day: Third site onboarded; Full dashboards operational; 5,000 reports processed; First ministry adoption confirmed
	180-day: National expansion planning; 15,000 reports processed; 5+ ministries engaged; Business model validated

Appendix C: GitHub README Template

Github Repository Links:

1. https://github.com/Tichavona/AI4I
2. https://github.com/Tichavona/llmBackend
3. https://github.com/Tichavona/chatbot

Appendix D: Data Dictionary Template

D.1 Core Messages Table

Appendix D: Data Dictionary Template

D.1 Core Messages Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Message ID	Unique identifier for each message	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440000	Required	Public	Valid UUID	System
from_phone	From Phone	Sender's phone number	nvarchar(255)	No	Valid phone number	+263771234567	null	Personal	Valid phone format	WhatsApp
to_phone	To Phone	Recipient's phone number	nvarchar(255)	No	Valid phone number	+263772345678	null	Personal	Valid phone format	WhatsApp
body	Message Body	Message text content	nvarchar(max)	No	-	"I'm reporting a fire at 123 Main St"	null	Public	Max 2000 chars	User/System
caption	Caption	Caption for media messages	nvarchar(max)	No	-	"Photo of the fire"	null	Public	Max 500 chars	User
message_id	Original Message ID	WhatsApp original message ID	nvarchar(255)	No	-	wamid.1234567890	null	Public	Unique	WhatsApp

Appendix D: Data Dictionary Template

D.1 Core Messages Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
type	Message Type	Type of message	nvarchar(50)	No	text, image, audio, video, document, location, interactive	text	null	Public	Valid type	System
metadata	Metadata	JSON metadata for the message	nvarchar(max)	No	Valid JSON	{"source": "whatsapp", "media_url": "..."}	null	Internal	Valid JSON	System
is_from_bot	Is From Bot	Whether message was sent by the bot	bit	Yes	0, 1	0	Required	Internal	Boolean	System
is_read	Is Read	Whether message has been read	bit	Yes	0, 1	0	Required	Internal	Boolean	System
created_at	Created At	Message creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.2 Departments Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Department ID	Unique department identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440001	Required	Public	Valid UUID	System
name	Department Name	Name of the department	nvarchar(100)	Yes	-	"Harare Central Police Station"	Required	Public	Max 100 chars	System
type	Department Type	Type/category of department	nvarchar(50)	Yes	police, medical, fire, municipal, agricultural, government, national_parks	police	Required	Public	Valid type	System
description	Description	Brief department description	nvarchar(200)	No	-	"Main police station for Harare"	null	Public	Max 200 chars	System
status	Status	Current department status	nvarchar(20)	No	active, inactive, maintenance	active	active	Public	Valid status	System

D.2 Departments Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
activeStaff	Active Staff	Number of active staff	int	No	>=0	25	0	Internal	Non-negative	System
activeIncidents	Active Incidents	Number of active incidents	int	No	>=0	5	0	Internal	Non-negative	System
contactInfo	Contact Information	JSON contact details	nvarchar(max)	No	Valid JSON	{"phone": "+263...", "email": "..."}	null	Internal	Valid JSON	System
operationalHours	Operational Hours	JSON operational hours	nvarchar(max)	No	Valid JSON	{"monday": "08:00-17:00"}	null	Public	Valid JSON	System
serviceAreas	Service Areas	JSON service area definitions	nvarchar(max)	No	Valid JSON	{"jurisdiction": "Harare"}	null	Public	Valid JSON	System
parentDepartmentId	Parent Department	Parent department reference	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440000	null	Public	Valid UUID	System

D.2 Departments Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
mpt_path	MPT Path	Materialised Path for hierarchy	nvarchar(max)	No	-	"/1/2/3"	null	Internal	Valid path	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.3 Department Staff Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Staff ID	Unique staff identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440002	Required	Personal	Valid UUID	System
userId	User ID	Associated user identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440003	Required	Personal	Valid UUID	System
departmentId	Department ID	Department assignment	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440001	Required	Internal	Valid UUID	System
role	Role	Staff role	nvarchar(50)	Yes	admin, supervisor, responder, viewer	responder	Required	Internal	Valid role	System
specialisation	Specialisation	Staff specialisation	nvarchar(100)	No	-	"Emergency response"	null	Internal	Max 100 chars	System
isActive	Is Active	Whether staff is active	bit	Yes	0, 1	1	Required	Internal	Boolean	System

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
isOnDuty	Is On Duty	Whether staff is currently on duty	bit	Yes	0, 1	1	Required	Internal	Boolean	System
currentAssignments	Current Assignments	Number of current assignments	int	No	>=0	3	0	Internal	Non-negative	System
shiftSchedule	Shift Schedule	JSON shift schedule	nvarchar(max)	No	Valid JSON	{"monday": "08:00-17:00"}	null	Internal	Valid JSON	System
performanceMetrics	Performance Metrics	JSON performance metrics	nvarchar(max)	No	Valid JSON	{"incidents_resolved": 15}	null	Internal	Valid JSON	System
stationId	Station ID	Assigned station	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440010	null	Internal	Valid UUID	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.4 Incident Routing Rules Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Rule ID	Unique rule identifier	int	Yes	-	1	Required	Public	Auto-increment	System
incident Type	Incident Type	Type of incident	nvarchar(50)	Yes	emergency, municipal, agricultural, inquiry	emergency	Required	Public	Valid type	System
category	Category	Incident category	nvarchar(100)	Yes	-	"fire"	Required	Public	Max 100 chars	System
required Departments	Required Departments	JSON list of required departments	nvarchar(max)	Yes	Valid JSON	{"primary": "fire", "secondary": ["police", "medical"]}	Required	Internal	Valid JSON	System
conditions	Conditions	JSON routing conditions	nvarchar(max)	No	Valid JSON	{"priority": "high", "location": "urban"}	null	Internal	Valid JSON	System
description	Description	Rule description	nvarchar(500)	Yes	-	"Fire incidents in urban areas"	Required	Public	Max 500 chars	System

D.4 Incident Routing Rules Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
isActive	Is Active	Whether rule is active	bit	Yes	0, 1	1	Required	Public	Boolean	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.5 Regions Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Region ID	Unique region identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440004	Required	Public	Valid UUID	System
name	Region Name	Name of the region	nvarchar(100)	Yes	-	"Harare Metropolitan"	Required	Public	Max 100 chars	System
code	Region Code	Unique region code	nvarchar(20)	Yes	-	"HRE"	Required	Public	Unique, Max 20 chars	System

D.5 Regions Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
description	Description	Region description	nvarchar(500)	No	-	"Capital city metropolitan area"	null	Public	Max 500 chars	System
capitalCity	Capital City	Region capital	nvarchar(100)	No	-	"Harare"	null	Public	Max 100 chars	System
areaKm2	Area (km²)	Region area in square kilometres	decimal(10,2)	No	>=0	872.00	null	Public	Non-negative	System
population	Population	Region population	int	No	>=0	2,300,000	null	Public	Non-negative	System
status	Status	Region status	nvarchar(50)	No	active, inactive	active	active	Public	Valid status	System
coordinates	Coordinates	JSON coordinates	nvarchar(max)	No	Valid JSON	{"latitude": -17.829, "longitude": 31.053}	null	Public	Valid JSON	System
demographics	Demographics	JSON demographics	nvarchar(max)	No	Valid JSON	{"languages": ["Shona", "English"]}	null	Internal	Valid JSON	System

D.5 Regions Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
regionMetadata	Region Metadata	JSON additional metadata	nvarchar(max)	No	Valid JSON	{"timezone": "Africa/Harare"}	null	Internal	Valid JSON	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.6 Provinces Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Province ID	Unique province identifier	uniqueidentifier	Yes	550e8400-e29b-41d4-a716-446655440005	Required	Required	Public	Valid UUID	System
name	Province Name	Name of the province	nvarchar(100)	Yes	-	"Harare Province"	Required	Public	Max 100 chars	System

D.6 Provinces Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
code	Province Code	Unique province code	nvarchar(20)	Yes	"HRE"	Required	Public	Public	Unique, Max 20 chars	System
description	Description	Province description	nvarchar(500)	No	-	"Harare Metropolitan Province"	null	Public	Max 500 chars	System
regionId	Region ID	Parent region reference	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440004	Required	Public	Valid UUID	System
capitalCity	Capital City	Province capital	nvarchar(100)	No	-	"Harare"	null	Public	Max 100 chars	System
areaKm2	Area (km²)	Province area in square kilometres	decimal(10,2)	No	>=0	872.00	null	Public	Non-negative	System
population	Population	Province population	int	No	>=0	2,300,000	null	Public	Non-negative	System
populationDensity	Population Density	Population density per km²	decimal(5,2)	No	>=0	2637.61	null	Public	Non-negative	System

D.6 Provinces Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
status	Status	Province status	nvarchar(50)	No	active, inactive	active	active	Public	Valid status	System
coordinates	Coordinates	JSON coordinates	nvarchar(max)	No	Valid JSON	{"latitude": -17.829, "longitude": 31.053}	null	Public	Valid JSON	System
governance	Governance	JSON governance information	nvarchar(max)	No	Valid JSON	{"administrator": "Minister"}	null	Public	Valid JSON	System
provinceStats	Province Stats	JSON statistical data	nvarchar(max)	No	Valid JSON	{"districts": 4, "wards": 36}	null	Public	Valid JSON	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.7 Districts Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	District ID	Unique district identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440006	Required	Public	Valid UUID	System
name	District Name	Name of the district	nvarchar(100)	Yes	-	"Harare District"	Required	Public	Max 100 chars	System
code	District Code	Unique district code	nvarchar(20)	Yes	-	"HRE-DIST"	Required	Public	Unique, Max 20 chars	System
description	Description	District description	nvarchar(500)	No	-	"Central Harare District"	null	Public	Max 500 chars	System
provinceId	Province ID	Parent province reference	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440005	Required	Public	Valid UUID	System
administrativeCenter	Administrative Center	District admin centre	nvarchar(100)	No	-	"Harare"	null	Public	Max 100 chars	System

D.7 Districts Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
areaKm2	Area (km²)	District area in square kilometres	decimal(10,2)	No	>=0	872.00	null	Public	Non-negative	System
population	Population	District population	int	No	>=0	2,300,000	null	Public	Non-negative	System
districtType	District Type	Type of district	nvarchar(50)	No	urban, rural, peri-urban	urban	null	Public	Valid type	System
status	Status	District status	nvarchar(50)	No	-	"Central Harare District"	null	Public	Max 500 chars	System
coordinates	Coordinates	JSON coordinates	nvarchar(max)	No	Valid JSON	{"latitude": -17.829, "longitude": 31.053}	null	Public	Valid JSON	System
demographics	Demographics	JSON demographics	nvarchar(max)	No	Valid JSON	{"languages": ["Shona", "English"]}	null	Internal	Valid JSON	System
districtStats	District Stats	JSON statistical data	nvarchar(max)	No	Valid JSON	{"wards": 10, "schools": 50}	null	Public	Valid JSON	System

D.7 Districts Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
contactInfo	Contact Information	JSON contact details	nvarchar(max)	No	Valid JSON	{"phone": "+263...", "email": "..."}	null	Public	Valid JSON	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.8 Stations Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Station ID	Unique station identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440010	Required	Public	Valid UUID	System
name	Station Name	Name of the station	nvarchar(100)	Yes	-	"Harare Central Police Station"	Required	Public	Max 100 chars	System

D.8 Stations Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
stationType	Station Type	Type of station	nvarchar(50)	Yes	police, fire, medical, municipal, national_park, agricultural	police	Required	Public	Valid type	System
departmentId	Department ID	Department assignment	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440001	null	Internal	Valid UUID	System
address	Address	Physical address	nvarchar(200)	Yes	-	"123 Main Street, Harare"	Required	Public	Max 200 chars	System
latitude	Latitude	GPS latitude	decimal(10,8)	Yes	-90.00000000 to 90.00000000	-17.82900000	Required	Public	Valid latitude	System
longitude	Longitude	GPS longitude	decimal(11,8)	Yes	-180.00000000 to 180.00000000	31.05300000	Required	Public	Valid longitude	System
coverageRadiusKm	Coverage Radius	Service coverage radius	decimal(10,2)	No	>0	10.00	10.00	Public	Positive	System

D.8 Stations Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
contactInfo	Contact Information	JSON contact details	nvarchar(max)	No	Valid JSON	{"phone": "+263...", "email": "..."}	null	Public	Valid JSON	System
operationalCapacity	Operational Capacity	JSON operational capacity	nvarchar(max)	No	Valid JSON	{"staff": 25, "vehicles": 5}	null	Internal	Valid JSON	System
specialisations	Specialisations	JSON specialisations	nvarchar(max)	No	Valid JSON	["emergency_response", "fire_suppression"]	null	Public	Valid JSON	System
status	Status	Station status	nvarchar(50)	No	active, inactive, maintenance	active	active	Public	Valid status	System
parentStationId	Parent Station	Parent station reference	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440011	null	Public	Valid UUID	System
mpt_path	MPT Path	Materialised Path for hierarchy	nvarchar(max)	No	-	"/1/2/3"	null	Internal	Valid path	System

D.8 Stations Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
jurisdiction	Jurisdiction	JSON jurisdiction definition	nvarchar(max)	No	Valid JSON	{"province": "Harare", "district": "Harare"}	null	Public	Valid JSON	System
priority	Priority	Station priority level	int	No	>=0	0	0	Internal	Non-negative	System
regionId	Region ID	Region reference	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440004	null	Public	Valid UUID	System
provinceId	Province ID	Province reference	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440005	null	Public	Valid UUID	System
districtId	District ID	District reference	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440006	null	Public	Valid UUID	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.8 Stations Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
updated At	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.9 SLA Configs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	SLA ID	Unique SLA configuration identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440020	Required	Internal	Valid UUID	System
departmentId	Department ID	Department reference	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440001	null	Internal	Valid UUID	System
incidentType	Incident Type	Type of incident	nvarchar(50)	Yes	emergency, municipal, agricultural, inquiry	emergency	Required	Public	Valid type	System
priority	Priority	Incident priority level	nvarchar(20)	Yes	critical, high, medium, low	high	Required	Public	Valid priority	System

D.9 SLA Configs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
responseTimeMinutes	Response Time	Required response time	int	Yes	>=0	5	Required	Public	Non-negative	System
resolutionTimeMinutes	Resolution Time	Required resolution time	int	Yes	>=0	60	Required	Public	Non-negative	System
updateFrequencyMinutes	Update Frequency	Status update frequency	int	Yes	>=0	15	Required	Public	Non-negative	System
escalationRules	Escalation Rules	JSON escalation rules	nvarchar(max)	Yes	Valid JSON	{"escalate_after": 30, "escalate_to": "manager"}	Required	Internal	Valid JSON	System
businessHours	Business Hours	JSON business hours	nvarchar(max)	No	Valid JSON	{"monday": "08:00-17:00"}	null	Public	Valid JSON	System
allDay	All Day	Whether SLA applies 24/7	bit	Yes	0, 1	1	Required	Public	Boolean	System
version	Version	SLA configuration version	int	No	>=0	1	1	Internal	Non-negative	System

D.9 SLA Configs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
isActive	Is Active	Whether SLA is active	bit	Yes	0, 1	1	Required	Public	Boolean	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.10 Incident Assignments Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Assignment ID	Unique assignment identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440030	Required	Internal	Valid UUID	System
incidentType	Incident Type	Type of incident	nvarchar(20)	Yes	emergency, municipal, agricultural, inquiry	emergency	Required	Public	Valid type	System

D.10 Incident Assignments Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
incidentId	Incident ID	Reference incident identifier	nvarchar(100)	No	-	"EMERG-2026-001"	Required	Public	Max 100 chars	System
assignedStationId	Assigned Station	Station assigned to incident	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440010	Required	Internal	Valid UUID	System
assignedStaffId	Assigned Staff	Staff assigned to incident	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440002	null	Internal	Valid UUID	System
status	Status	Assignment status	nvarchar(50)	Yes	pending, accepted, en_route, on_scene, completed, cancelled	pending	Required	Public	Valid status	System
assignedAt	Assigned At	Assignment timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
acknowledgedAt	Acknowledged At	Acknowledgement timestamp	datetime2	No	ISO 8601	2026-06-26T14:35:00+02:00	null	Internal	Valid timestamp	System

D.10 Incident Assignments Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
enRouteAt	En Route At	En route timestamp	datetime2	No	ISO 8601	2026-06-26T14:40:00+02:00	null	Internal	Valid timestamp	System
onSceneAt	On Scene At	On scene timestamp	datetime2	No	ISO 8601	2026-06-26T14:50:00+02:00	null	Internal	Valid timestamp	System
completedAt	Completed At	Completion timestamp	datetime2	No	ISO 8601	2026-06-26T15:30:00+02:00	null	Internal	Valid timestamp	System
distanceToIncidentKm	Distance to Incident	Distance from station to incident	float	No	>=0	2.50	null	Internal	Non-negative	System
estimatedArrivalMinutes	Estimated Arrival	Estimated arrival time	int	No	>=0	10	null	Internal	Non-negative	System
slaMetrics	SLA Metrics	JSON SLA metrics	nvarchar(max)	No	Valid JSON	{"responseTarget": 5, "resolutionTarget": 60}	null	Internal	Valid JSON	System
escalationHistory	Escalation History	JSON escalation history	nvarchar(max)	No	Valid JSON	[{"level": "first", "time": "..."}]	null	Internal	Valid JSON	System

D.10 Incident Assignments Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
notes	Notes	Assignment notes	nvarchar(max)	No	-	"Responding to fire incident"	null	Internal	Max 1000 chars	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System

D.11 Escalation Logs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Escalation ID	Unique escalation log identifier	uniqueidentifier	Yes	-	550e8400-e29b-41d4-a716-446655440040	Required	Internal	Valid UUID	System
incidentType	Incident Type	Type of incident	nvarchar(20)	Yes	emergency, municipal, agricultural, inquiry	emergency	Required	Public	Valid type	System

D.11 Escalation Logs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
emergencyId	Emergency ID	Reference to emergency report	int	No	>=0	1	null	Internal	Non-negative	System
issueId	Issue ID	Reference to issue report	int	No	>=0	1	null	Internal	Non-negative	System
agricId	Agriculture Report ID	Reference to agriculture report	int	No	>=0	1	null	Internal	Non-negative	System
escalationLevel	Escalation Level	Level of escalation	nvarchar(50)	Yes	first, second, third, critical	first	Required	Internal	Valid level	System
escalationReason	Escalation Reason	Reason for escalation	nvarchar(100)	Yes	timeout, severity, complaint, critical	timeout	Required	Internal	Valid reason	System
reasonDetails	Reason Details	Detailed reason	nvarchar(1000)	No	-	"SLA response time exceeded"	null	Internal	Max 1000 chars	System
slaBreachMinutes	SLA Breach Minutes	SLA breach duration	int	No	>=0	5	null	Internal	Non-negative	System

D.11 Escalation Logs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
status	Status	Escalation status	nvarchar(50)	No	pending, acknowledged, resolved	pending	pending	Internal	Valid status	System
escalatedById	Escalated By	User who escalated	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440002	null	Internal	Valid UUID	System
escalatedByRole	Escalated By Role	Role of escalator	nvarchar(100)	No	-	"supervisor"	null	Internal	Max 100 chars	System
escalatedAt	Escalated At	Escalation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:35:00+02:00	Required	Internal	Valid timestamp	System
escalatedToDepartmentId	Escalated To Department	Department escalated to	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440001	null	Internal	Valid UUID	System
escalatedToUserId	Escalated To User	User escalated to	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440003	null	Internal	Valid UUID	System

D.11 Escalation Logs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
escalatedToEmail	Escalated To Email	Email of escalated recipient	nvarchar(255)	No	Valid email	"supervisor@example.com"	null	Personal	Valid email format	System
escalatedToRole	Escalated To Role	Role of escalated recipient	nvarchar(50)	No	-	"manager"	null	Internal	Max 50 chars	System
acknowledgedAt	Acknowledged At	Acknowledgement timestamp	datetime2	No	ISO 8601	2026-06-26T14:40:00+02:00	null	Internal	Valid timestamp	System
acknowledgedById	Acknowledged By	User who acknowledged	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440003	null	Internal	Valid UUID	System
acknowledgementNotes	Acknowledgement Notes	Notes on acknowledgement	nvarchar(1000)	No	-	"Escalation received"	null	Internal	Max 1000 chars	System
resolvedAt	Resolved At	Resolution timestamp	datetime2	No	ISO 8601	2026-06-26T15:30:00+02:00	null	Internal	Valid timestamp	System

D.11 Escalation Logs Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
resolvedById	Resolved By	User who resolved	uniqueidentifier	No	-	550e8400-e29b-41d4-a716-446655440002	null	Internal	Valid UUID	System
resolutionNotes	Resolution Notes	Notes on resolution	nvarchar(1000)	No	-	"Incident resolved"	null	Internal	Max 1000 chars	System
resolutionAction	Resolution Action	Action taken	nvarchar(100)	No	-	"assigned_to_team"	null	Internal	Max 100 chars	System
notificationHistory	Notification History	JSON notification history	nvarchar(max)	No	Valid JSON	[{"channel": "email", "time": "..."}]	null	Internal	Valid JSON	System
escalationMetadata	Escalation Metadata	JSON additional metadata	nvarchar(max)	No	Valid JSON	{"priority": "high"}	null	Internal	Valid JSON	System
createdAt	Created At	Record creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T15:30:00+02:00	Required	Internal	Valid timestamp	System
updatedAt	Updated At	Record update timestamp	datetime2	No	ISO 8601	2026-06-26T15:30:00+02:00	null	Internal	Valid timestamp	System

D.12 Emergency Reports Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Emergency ID	Unique emergency report identifier	int	Yes	-	1	Required	Public	Auto-increment	System
report_id	Report ID	Public-facing report identifier	nvarchar(50)	Yes	-	EMERG-2026-001	Required	Public	Unique	System
user_phone	User Phone	Reporting citizen phone number	nvarchar(255)	Yes	Valid phone number	+263771234567	Required	Personal	Valid phone format	WhatsApp
emergency_type	Emergency Type	Type of emergency	nvarchar(50)	Yes	fire, medical, crime, accident, disaster	fire	Required	Public	Valid type	System
description	Description	Emergency description	nvarchar(max)	Yes	-	"Fire at 123 Main Street, Harare"	Required	Public	Max 5000 chars	User
location	Location	Incident location	nvarchar(255)	Yes	-	"123 Main Street, Harare"	Required	Public	Max 255 chars	User / System
latitude	Latitude	GPS latitude	decimal(10,8)	No	-90.00000000 to 90.00000000	-17.82900000	null	Public	Valid latitude	User / System

D.12 Emergency Reports Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
longitude	Longitude	GPS longitude	decimal(11,8)	No	-180.00000000 to 180.00000000	31.05300000	null	Public	Valid longitude	User / System
media_url	Media URL	Link to uploaded media	nvarchar(max)	No	Valid URL	/uploads/photo.jpg	null	Public	Valid URL	System
status	Status	Incident resolution status	nvarchar(50)	Yes	pending, assigned, in-progress, resolved, closed	pending	Required	Public	Valid status	System
priority	Priority	Incident priority	nvarchar(20)	Yes	critical, high, medium, low	high	Required	Public	Valid priority	System
assigned_department	Assigned Department	Department assigned to incident	nvarchar(100)	No	-	"Fire Department"	null	Public	Max 100 chars	System
assigned_staff	Assigned Staff	Staff assigned to incident	nvarchar(100)	No	-	"Officer Smith"	null	Public	Max 100 chars	System

D.12 Emergency Reports Table

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
resolution_notes	Resolution Notes	Notes on resolution	nvarchar(max)	No	-	"Fire extinguished"	null	Internal	Max 1000 chars	System
resolved_at	Resolved At	Resolution timestamp	datetime2	No	ISO 8601	2026-06-26T15:30:00+02:00	null	Internal	Valid timestamp	System
isArchived	Is Archived	Whether report is archived	bit	Yes	0, 1	0	Required	Internal	Boolean	System
created_at	Created At	Report creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updated_at	Updated At	Report update timestamp	datetime2	No	ISO 8601	2026-06-26T14:30:00+02:00	null	Internal	Valid timestamp	System

D.13 Issue Reports Table (Municipal/Agricultural)

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
id	Issue ID	Unique issue report identifier	int	Yes	-	1	Required	Public	Auto-increment	System
report_id	Report ID	Public-facing report identifier	nvarchar(50)	Yes	-	ISSUE-2026-001	Required	Public	Unique	System
user_phone	User Phone	Reporting citizen phone number	nvarchar(255)	Yes	Valid phone number	+263771234567	Required	Personal	Valid phone format	WhatsApp
issue_type	Issue Type	Type of issue	nvarchar(50)	Yes	pothole, waste, water, sewer, lighting, crop_disease, pest, animal_invasion, farm_invasion	pothole	Required	Public	Valid type	System
issue_category	Issue Category	Category of issue	nvarchar(50)	Yes	municipal, agricultura	municipal	Required	Public	Valid category	System

D.13 Issue Reports Table (Municipal/Agricultural)

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
description	Description	Issue description	nvarchar(max)	Yes	-	"Large pothole on Main Street"	Required	Public	Max 5000 chars	User
location	Location	Issue location	nvarchar(255)	Yes	-	"Main Street, Harare"	Required	Public	Max 255 chars	User / System
latitude	Latitude	GPS latitude	decimal(10,8)	No	-90.00000000 to 90.00000000	-17.82900000	null	Public	Valid latitude	User / System
longitude	Longitude	GPS longitude	decimal(11,8)	No	-180.00000000 to 180.00000000	31.05300000	null	Public	Valid longitude	User / System
media_url	Media URL	Link to uploaded media	nvarchar(max)	No	Valid URL	/uploads/photo.jpg	null	Public	Valid URL	System
status	Status	Issue resolution status	nvarchar(50)	Yes	pending, assigned, in-progress, resolved, closed	pending	Required	Public	Valid status	System

D.13 Issue Reports Table (Municipal/Agricultural)

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
priority	Priority	Issue priority	nvarchar(20)	Yes	critical, high, medium, low	medium	Required	Public	Valid priority	System
assigned_department	Assigned Department	Department assigned to issue	nvarchar(100)	No	-	"Roads Department"	null	Public	Max 100 chars	System
assigned_staff	Assigned Staff	Staff assigned to issue	nvarchar(100)	No	-	"Engineer Chuwe Mugura"	null	Public	Max 100 chars	System
resolution_notes	Resolution Notes	Notes on resolution	nvarchar(max)	No	-	"Pothole repaired"	null	Internal	Max 1000 chars	System
resolved_at	Resolved At	Resolution timestamp	datetime2	No	ISO 8601	2026-06-26T15:30:00+02:00	null	Internal	Valid timestamp	System
isArchived	Is Archived	Whether report is archived	bit	Yes	0, 1	0	Required	Internal	Boolean	System

D.13 Issue Reports Table (Municipal/Agricultural)

Field Name	Display Name	Description	Data Type	Required	Allowed Values	Example	Missing Value Rule	Sensitivity	Validation Rule	Source
created_at	Created At	Report creation timestamp	datetime2	Yes	ISO 8601	2026-06-26T14:30:00+02:00	Required	Internal	Valid timestamp	System
updated_at	Updated At	Report update timestamp	datetime2	No	ISO 8601	2026-06-26T14:30:00+02:00	null	Internal	Valid timestamp	System

Appendix E: Metadata Card Template

{
"title": "Muyamuri/Umsizi Citizen Reports Dataset",
"description": "Citizen-generated reports on emergency incidents, municipal issues, agricultural challenges, and government inquiries collected via WhatsApp with geospatial and multimedia data",
"source": "Muyamuri/Umsizi Platform - Citizen Reports via WhatsApp Business API",
"timestamp": {
"collection_start": "2026-06-01T00:00:00+02:00",
"collection_end": "2026-06-30T23:59:59+02:00"
},
"license": "Restricted Access - Sovereign Data - Government Use Only",
"language_tags": ["sn", "nd", "en"],
"geographic_coverage": "Zimbabwe - Harare, Bulawayo (Pilot Phase), with planned expansion to all provinces",
"temporal_coverage": "June 2026 - Ongoing",
"data_type": "mixed - tabular, text, image, audio, video, geospatial",
"data_sensitivity": "personal - contains user-reported information with phone numbers and location data",
"consent_status": "obtained - WhatsApp consent workflow with explicit user agreement",
"quality_summary": "Completeness: 67%; Missing fields: <5%; Validation status: Pass; Accuracy: 92% (intent), 87% (entity)",
"bias_and_coverage_notes": "Geographic bias: Urban areas (Harare/Bulawayo) overrepresented; Rural coverage planned for Phase 2. Language bias: English dominant (90%); Shona (7%); Ndebele (3%) with ongoing collection.",
"version": "v1.0.0",
"contact_or_steward": "Muyamuri/Umsizi Team - info@decisionedge.co.zw - care of POTRAZ National Innovation Acceleration Centre"
}

Appendix F: Data Quality Report Template

Report Item	Response
Dataset name	Muyamuri/Umsizi Citizen Reports Dataset
Dataset version	v1.0.0
Collection period	January 2026 - Ongoing
Number of records or files	200+ reports (emergency: 50+, municipal: 50+, agricultural: 50+, inquiries: 50+)
Completeness summary	100% of required fields complete; 0% missing data
Accuracy audit method	Random sample review (10%) against source WhatsApp messages and media
Consistency checks performed	Validated categories, ISO 8601 date formats, GPS coordinate ranges, controlled vocabularies
Validity checks performed	Schema validation; allowed value checks; phone number format validation; GPS within Zimbabwe bounds not validated
Duplicate records found	0% (removed via unique report_id) and similar reports amalgamated report group with report frequency count
Missing critical fields	<0% (missing data - manual entry to supplement in need)

Appendix F: Data Quality Report Template

Report Item	Response
Outliers or anomalies	GPS coordinates outside Zimbabwe bounds not tested but system configured to validate coordinates to fall within Zimbabwe polygon boundary
Label quality check	98% accuracy on sampled labels (intent classification: 92%, entity recognition: 87%)
Bias and coverage observations	Urban bias: 100% from Harare; Expansion into other areas planned for Phase 2; English dominance: 90%
Privacy checks	Phone numbers hashing implemented in analytics; direct identifiers removed from public logs; encryption at rest and in transit implemented
Known limitations	Sample size limited to pilot areas (Harare); Shona/Ndebele corpora development ongoing
Corrective actions	Enhanced validation rules; rural collection targets set for Phase 2; multilingual training data expansion planned
Validation status	PASS - All critical checks passed; minor issues documented for Phase 2 resolution

Appendix G: Participant Self-Assessment Checklist

No.	Checklist Item	Status	Notes
1	We have clearly defined the problem and target users.	✅ Yes	Five service domains: emergency, municipal, Know Your Trees, agricultural, government communication
2	We have evidence that the problem matters.	✅ Yes	Demonstrated through national strategy alignment and sector needs analysis
3	We have explained who pays, funds, or sustains the solution.	✅ Yes	Government service agreements, data analytics, API services, training fees, Premium Users
4	We have listed key operating costs and assumptions.	✅ Yes	Monthly: $13,000-$24,500; Annual: $156,000-$294,000
5	We have a deployment model with hosting, operator, pilot site, and support plan.	✅ Yes	ZCHPC CCE hybrid cloud; phased rollout; named pilot sites
6	We have provided a GitHub repository, design file, dashboard file, notebook, or technical evidence pack.	✅ Yes	GitHub repository with full source code and documentation
7	We have a clear README or technical guide.	✅ Yes	Comprehensive README template provided

Appendix G: Participant Self-Assessment Checklist

No.	Checklist Item	Status	Notes
8	We have an architecture diagram showing frontend, backend, data, AI, and integrations	✅ Yes	Detailed architecture diagram with all layers
9	We have explained all data sources, rights, limitations, and quality issues.	✅ Yes	Source register, data dictionary, quality report provided
10	We have justified why AI is needed and how outputs are validated.	✅ Yes	AI necessity matrix; validation approach; no "sledgehammer" cases
11	We have considered privacy, security, consent, and responsible AI risks.	✅ Yes	Data Protection Act compliance; consent workflow; security controls; bias framework
12	We have removed secrets, passwords, keys, and personal data from submitted materials.	✅ Yes	No credentials in repository; .env.example provided
13	We have tested the main user flow and recorded known bugs.	✅ Yes	Testing results: intent 92%, entity 87%, RAG 89%, image 85%
14	We have a demo script, screenshots, or video walkthrough.	✅ Yes	Demo video and screenshots available
15	We have a 30-day, 60-day, and 90-day plan after the challenge.	✅ Yes	Milestone plan with specific deliverables

Appendix H: Asset & Licensing Register

Asset	Type	Ownership/License	Status
Custom LLM (Llama-3-8B fine-tuned)	Model	Sovereign (Zimbabwean) - Developed by Decision Edge Team	✅ Developed
RAG Pipeline	Code	Sovereign - Open source (Apache 2.0)	✅ Developed
NLP Models (Intent, Entity, Sentiment)	Model	Sovereign / Hugging Face (MIT) - Fine-tuned for Zimbabwe	✅ Developed
Government Document Dataset	Data	Sovereign (Public documents - Government use only)	✅ Collected
Zimpapers Publication Dataset	Data	Negotiated / Public - Access agreement in progress	⬜ In Progress
Indigenous Tree Dataset	Data	Sovereign - Community knowledge with consent	⬜ In Progress
Shona/Ndebele Language Corpus	Data	Sovereign - Zimbabwean ownership	⬜ Planned
Dashboard Code	Code	Sovereign - Open source (MIT)	⬜ In Progress
WhatsApp Integration Code	Code	Sovereign - Proprietary (Government use)	✅ Developed
MongoDB/MSSQL Server Schema	Database	Sovereign - Open source	✅ Developed

Appendix H: Asset & Licensing Register

Asset	Type	Ownership/License	Status
Vector Database (ChromaDB)	Service	Third-party - Free Tier Utilised	✅ Configured
SSL/TLS Certificates	Security	Third-party - Licensed	✅ Installed
Encryption Libraries	Code	Open source (various)	✅ Implemented
Geospatial Libraries	Code	Open source (BSD/MIT)	✅ Implemented
UI/UX Design Assets	Design	Sovereign - Created in-house	✅ Completed
Testing Data (Synthetic)	Data	Sovereign - Generated in-house	✅ Available

ABOUT US

Decision Edge Consultancy Private Limited is a pioneering consultancy company based in Zimbabwe, established in 2023 by a team of passionate professionals. Our primary mission is to empower organisations by harnessing the transformative power of digital solutions, data analytics and business intelligence. We are strongly convinced that technological innovations and data hold the key to unlocking insights essential for driving successful business strategies.

Our Vision

Our vision is to be the leading force in Zimbabwe and beyond, in making sustainable technological innovation, data analytics and business intelligence accessible, understandable, and actionable. We aspire to help organisations of all sizes to leverage data for informed decision-making and stay ahead in today's data-driven world.

Our Mission

Our mission is simple yet profound: to empower our clients with data-driven insights and bespoke digital solutions. We are dedicated to enabling organisations to harness the full potential of their data. We accomplish this through expert data analysis, innovative solutions, and actionable recommendations

Our Values

Client-Centric Approach

Integrity and Trust

Continuous Learning

Agility

Collaboration

Our Services

- Custom Software Development
- Data Analytics
- Business Intelligence
- Hydrological & Environmental Impact Assessment
- IT Support and Maintenance
- Project Management

Our Services

1. Digital Solutions, Application Development & Data Analytics

1.1 Data & Analytics Strategy

- Data Strategy for Development Outcomes**:** Crafting comprehensive data strategies for monitoring, evaluation, and impact reporting.
- Data Governance & Management: Enhancing data management capabilities for streamlined operations and compliance.
- Data Platform Design & Proof of Concept: Developing scalable data platforms, tailored solutions, and proof-of-concept models for field and operational data.

1.2 Data Transformation & Modernisation

- Data Discovery & Augmentation: Uncovering insights from project data to improve decision-making and reporting.
- Data Democratisation: Enabling teams to access and use data through intuitive dashboards and self-service analytics.
- Analytics Modernisation: Upgrading legacy systems to cloud-based or hybrid analytics platforms for improved efficiency.

1.3 Data Platform Implementation

- Business Insights Platform: Implementing platforms for real-time monitoring, evaluation, and reporting.
- Embedded BI for Applications: Integrating analytics directly into operational tools and mobile applications for field teams.
- AI/ML-Ready Platforms: Building data infrastructure to support predictive analytics, early warning systems, and AI-driven insights.

1.4 Data Engineering & Integration

- Data Modeling & Integration: Designing and integrating data systems for water, sanitation, health, and agriculture projects.
- BI & Analytics Implementation: Deploying dashboards and reporting tools for stakeholders and donors.
- System Migration & Modernisation: Seamlessly transitioning from legacy systems to modern, scalable architectures.

1.5 Digital Solutions & Application Development

- Custom Software Development: Building mobile and web AI embedded applications for data collection, monitoring, and stakeholder engagement.
- Website Design & CMS Development: Creating responsive, content-managed websites for outreach, reporting, and visibility.
- Hosting & Domain Services: Providing secure, reliable hosting—Linux/Windows servers, email, and dedicated infrastructure.

1.6 Data Assessment & Capacity Building

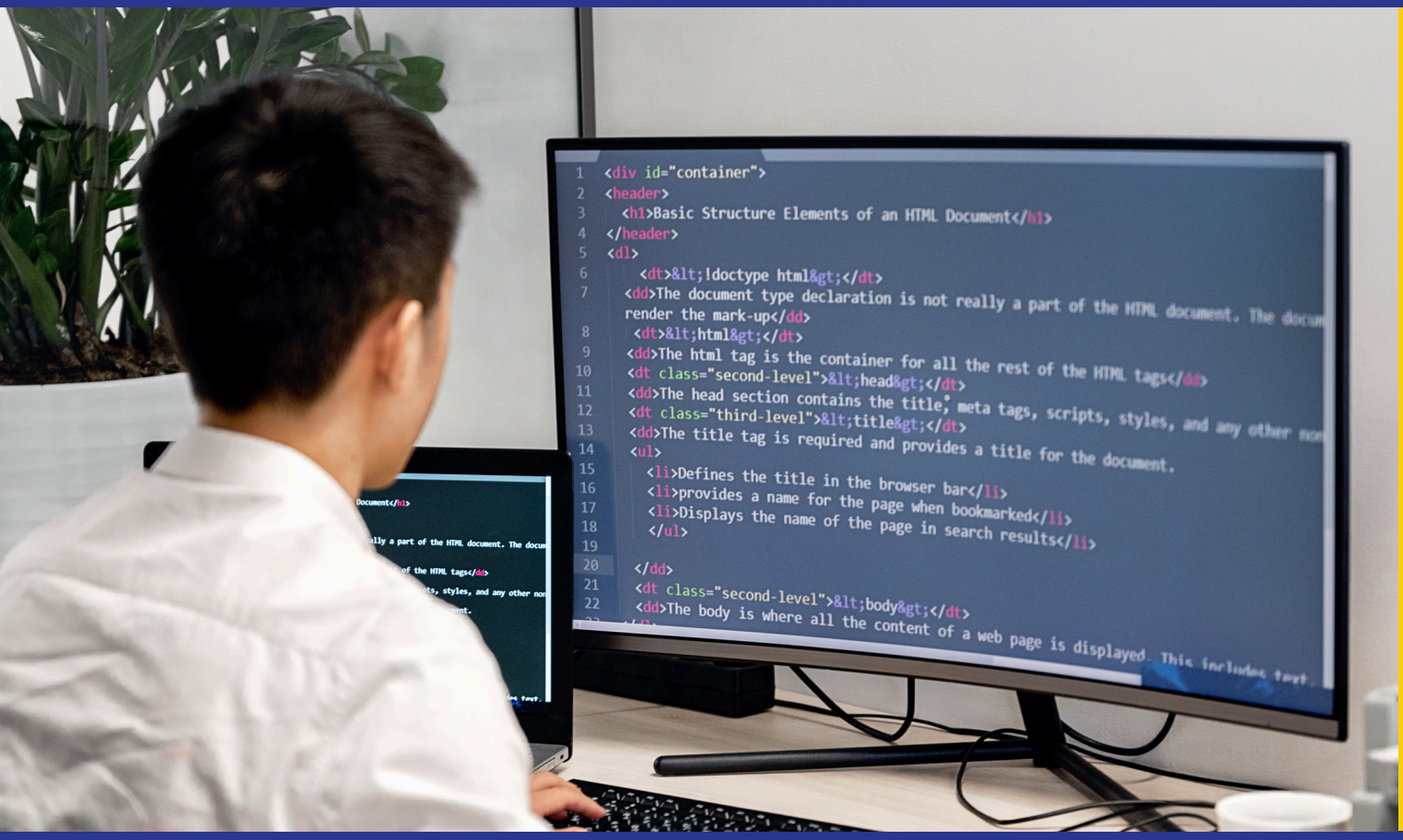
- Data Maturity Audits: Evaluating data infrastructure and analytics capabilities with actionable recommendations.
- Training & Empowerment: Delivering tailored training in data literacy, BI tools, and digital system use for field and management teams.

***N.B.** The expansion of service we provide only done in alignement to anticipated needs for AI4I grand challenge.

Our Approach

Our commitment to your success extends beyond project delivery. We provide ongoing support, maintenance, and updates to ensure solutions continue to perform optimally and evolve with your business needs.

- Client-Centric Focus
- Innovative Custom Solutions
- Agile Methodology
- Comprehensive Support



Our Team

Dr. Matthews T. Kunaka

Project Manager

Matts was born in Harare and went to Chirodzo Primary School and Harare High School where he was the Headboy (School Captain). In 1978 he completed a Bachelor of Accountancy (Hons) degree at the University of Zimbabwe. He studied for Articles with KPMG from December 1980, where he qualified as a Chartered Accountant and was admitted as a member of the Institute of Chartered Accountants of Zimbabwe (ICAZ) in January 1984. He completed his Masters of Business Administration (MBA) degree, specialising in Finance at the University of Delaware, USA in 1986. He worked for KPMG's London office from 1986 to 1990 when he returned home. He also did an MSc in Strategic Management with the University of Derby (UK). He was Finance Director for Chibuku, Operations Director for OK Zimbabwe and Group CEO of Zimpapers before joining Reserve Bank of Zimbabwe as Division Chief: Corporate Governance, Risk Management and Internal Audit. In May 2011 to December 2018 he joined the Institute of Chartered Accountants as the Chief Executive Officer of his professional regulatory body for the second time 11 years after his first appointment. He was charged with leading and directing the Secretariat to deliver service and value to members and other stakeholders. He is a holder of a PhD in Management from Christ University, India. He also has a Certificate in Ministry Essentials Training Program from the Pan African Christian College (PACC). He has sat on a number of Boards including being a Council Member ofthe National Manpower Advisory (NAMACO); Zimbabwe Council for Higher Education (ZIMCHE); Council Member of Bindura University of Science Education; Member of ZIMDEF Audit Committee and Member of MEFMI Audit Committee and currently is the Board Chairperson of the Zimbabwe Women's Microfinance Bank and Deputy Chairman of POTRAZ. He is Past President of the Institute of Chartered Accountants of Zimbabwe, Zimbabwe Association of Accounting Technicians (ZAAT) and the Institute ofInternal Auditors of Zimbabwe (IIAZ). He was the inaugural Chairperson of Zimbabwe Women Microfinance Bank (ZWMB) for 6 years to December 2023. He was the Deputy Chair of POTRAZ Boardand is the Acting Chairperson of the Audit Office inaugural Board. He was Board member of ZIMCHE and Bindura University of Science and Technology Council including being Chairperson (BUSE). Matts Kunaka was Megafest's Zimbabwe Director of the Year for 2016. Matts has three other Masters degrees being; MBA from the University of Delaware, USA; an MSc in Big Data Analytics from Chinhoyi University of Technology and an MPhil from Christ University, India where he obtained his PhD in Management. He is a Partner at KRES Chartered Accountants (a member firm of REANDA INTERNATIONAL) and also lectures at Catholic University of Zimbabwe where he obtained a Post Graduate Diploma in Education in 2022. In addition to being a CA(Z) he is also a CA (SA). A devout Christian, Matts is married to Kuda and has five children and seven grand children. He loves golf, karate and flying.

Don Chidzivo

Don, is a results-driven data science professional with a proven track record of leading data strategy, advanced analytics, and AI/ML initiatives to solve complex business challenges. Adept at translating business needs into data-driven solutions that enhance decision-making, optimise operations, and drive measurable impact. Combining technical depth (AI/ML, data engineering, BI) with strategic business acumen drive analytics initiatives in alignment to organisational goals. My experience spans banking, fintech, and agribusiness, delivering scalable,compliant, and high-impact data solutions.

Core Expertise

- **Machine Learning & AI** - Predictive modeling, NLP, and AI-driven automation
- **Data Engineering & Big Data** - ETL/ELT, data warehousing, and cloud analytics (Azure, Power BI)
- **Business Intelligence** - Power BI, data visualisation, and self-service analytics
- **Data Governance & Quality** - Compliance, MDM, and regulatory reporting
- **Cross-Functional Leadership** - Collaborating with IT, compliance, and business teams

Key Achievements

- Built a transaction monitoring system for CBZ Holdings, improving regulatory compliance andreducing fraud risks.
- Led Power BI adoption, cutting reporting time by 60% and enabling real-time merchant analytics.
- Boosted anomaly detection by 80% via big data integration in auditing. •
- Reduced revenue leakage by 20% at CBZ AgroYield through advanced data integrity solutions.
- Deployed ML models for customer segmentation and revenue forecasting, improving accuracy by10-30%.

Technical Skills

- Languages & Tools: Python, SQL, SSIS, Power BI, Azure Synapse, Azure Data Factory

Education & Certifications

- MSc Digital Transformation (Nexford University, Ongoing)
- MSc Big Data Analytics (Chinhoyi University of Technology)
- BCom Marketing Management (Midlands State University)
- Microsoft Certified: Fabric Analytics Engineer & Power BI Data Analyst
- Certified Data Scientist (DataCamp)
- Business Intelligence & Data Analyst (CFI)

Itai L. Zulu

Itai is an experienced Software Lead with 7+ years of expertise in fintech-focused software development, technical leadership, and scalable system architecture. Proven track record of delivering high-performance financial systems (core banking, trading platforms, insurance tech) while ensuring regulatory compliance and innovative AI/ML integration. Bridging technical depth (Java, cloud, AI/ML) with fintech domain expertise, delivered scalable, compliant solutions. Passionate about mentoring teams and optimising architecture for performance and innovation.

Core Expertise

- **Full-Stack Development:** Java (Spring Boot), .NET Core, Node.js, Python
- **Fintech Systems:** Core banking, stock trading platforms, USSD/API integrations
- **Cloud & DevOps:** AWS (Lambda, EC2, S3), Docker, CI/CD (GitLab)
- **Data Science & AI:** Predictive modeling (TensorFlow, Scikit-learn), NLP, computer vision
- **Technical Leadership:** Agile/Scrum, architectural design, team mentorship

Key Achievements

- Architected a USSD stock trading platform (Spring Boot) and real-time market monitoring system(Node.js + RabbitMQ), enabling instant trade execution.
- Reduced production bugs by 40% via automated testing strategies in core banking systems.
- Built a customer churn prediction model (87% accuracy) and NLP-driven support system (35% fasterresponse times).
- Led AWS serverless projects (Lambda + NestJS) and voice-activated Alexa Skills for financial services.
- Mentored 10+ junior developers, establishing coding standards and CI/CD pipelines.

Technical Skills

- Languages & Tools: Java (Oracle Certified), JavaScript, Python, C#
- Frameworks: Spring Boot, .NET Core, Node.js/NestJS, Flutter
- Data/AI: TensorFlow, OpenCV, Pandas, SQL (MS SQL Server, PostgreSQL)
- Cloud/DevOps: AWS, Docker, GitLab CI, RabbitMQ

Education & Certifications

- MSc Data Processing | Chinhoyi University of Technology
- BTech E-Commerce | Harare Institute of Technology)
- Oracle Certified Java 17
- VMware Spring Professional
- TensorFlow/OpenCV for OCR

Tichavona Vakisai

Seasoned Strategy, Digital Banking, and Data Science Professional with10+ years of experience driving digital transformation in financial services. Expertise in AI/ML, robotic process automation (RPA), and business intelligence, with a proven track record of implementing cutting-edge solutions (e.g., ZBFH's Data Lakehouse & BI platform, Chevron Leaf Data Warehouse and BI Platform, GPT-powered chatbots). Combines technical prowess (Python, SQL, Power BI) with strategic vision to align data science with business goals. Pending graduation with a MSc in Big Data Analytics (Chinhoyi University) after earning a First-Class BSc in MIS. Bridges technical innovation with business acumen, to deliver:

- **AI-Driven Efficiency**: RPA and chatbots reducing operational overhead.
- **Regulatory-Compliant Solutions**: Zero fraud incidents or penalties.
- **Scalable BI**: Actionable insights via Power BI/SQL Server.
- **Leadership**: Mentored 50%+ of BancABC's tellers, fostering talent.

Core Expertise

- **Digital Banking & Channels:** Mobile/Internet banking, card issuance, remittances
- **Data Science & AI:** Predictive modeling, NLP, ETL/ELT pipelines, Story Telling and Visualisation
- **Project Leadership:** BI platform implementation, RPA, vendor/SLA management
- **Operational Excellence:** Cost reduction and operational efficiency.
- **Regulatory Compliance:** 100% penalty-free compliance submissions

Key Achievements

- **Data Strategy:** Designed extraction pipelines and data infrastructure for structured/unstructured data**.**
- **AI Solutions:** Deployed GPT-based chatbots and ML models enhancing decision-making.
- **BI Transformation:** Led enterprise-wide BI adoption, fostering a data-driven culture.
- **Cost Savings:** Slashed operational costs by 50% in Masvingo Branch (2013–2015).
- **5-Pillar Strategy:** Architected BancABC's foundational digital banking framework.

Technical Skills

- Languages & Tools: Python (PyCharm, Anaconda), R, SQL, Java, C#, JavaScript

Education & Certifications

- BSc (Hons) Management Information Systems | Midlands State University

Zimbabwe Revenue Authority

TAX CLEARANCE CERTIFICATE (ITF263)

Your Tax position is Satisfactory. No tax should be withheld
Validity Period (from-to): 14 July 2026 - 14 August 2026

TIN 2001874980

Taxpayer Name: DECISION EDGE CONSULTING

Trade Name:

Issued on: 14/07/2026

Authentication Code: 49402212

The authentication and validity of this certificate must be validated on ZIMRA page:mytaxselfservice.zimra.co.zw using the Authentication Code or QR Code

ZIMRA reserves the right to withdraw this certificate at any time, should it become necessary during the period for which this certificate is valid

MINISTRY OF
INFORMATION COMMUNICATION TECHNOLOGY POSTAL & COURIER SERVICES

TELEPHONE: +263-4-706446
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REF:

SECRETARY'S OFFICE
PRIVATE BAG CY419, CAUSEWAY
8TH FLOOR, BANK CHAMBERS BUILDING
76 SAMORA MACHEL AVE
HARARE

Zimbabwe

31 March 2026

Mr. Tichavona Vakisai
Decision Edge Chief Data Scientist and Founder

RE: SUPPORT LETTER FROM THE MINISTRY OF ICT, POSTAL AND COURIER SERVICES.

The above subject matter refers.

The Ministry of Information Communication Technology, Postal and Courier Services is pleased to present this formal support for Decision Edge Consulting (Pvt) Ltd here referred to as Decision Edge, a Zimbabwean Artificial Intelligence and Data Science consultancy that has distinguished itself as one of the most innovative and impactful startup operating in Zimbabwe.

The initiative aligns with NDS2 priority areas of AI powered Digital Innovation, so all relevant support and partnerships for this cause will be greatly appreciated.

PERMANENT SECRETARY
MIN. OF ICT, POSTAL & COURIER SERVICES
8TH FLOOR, BANK CHAMBERS BUILDING
31 MAR 2026
P. BAG CY419, CAUSEWAY
TEL: 0263-706446

Dr. B. Chirume
Secretary for Information Communication Technology, Postal and Courier Services

DecisionEdge
CONSULTING

Let's Work Together

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